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# Earth and Space Science Performance Targets

for the  
South Carolina College- and Career-Ready Science Standards 2021

For use 2026-2027

July 2026

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## Contents

|                                            |    |
|--------------------------------------------|----|
| Purpose and Use.....                       | 4  |
| Document Updates.....                      | 5  |
| July 2026 .....                            | 5  |
| July 2025 .....                            | 5  |
| June 2024 .....                            | 5  |
| ESS1 – Earth’s Place in the Universe ..... | 6  |
| E-ESS1-1.....                              | 6  |
| E-ESS1-1 Academic Language .....           | 8  |
| E-ESS1-2.....                              | 9  |
| E-ESS1-2 Academic Language .....           | 11 |
| E-ESS1-3.....                              | 12 |
| E-ESS1-3 Academic Language .....           | 14 |
| E-ESS1-4.....                              | 15 |
| E-ESS1-4 Academic Language .....           | 17 |
| E-ESS1-5.....                              | 18 |
| E-ESS1-5 Academic Language .....           | 20 |
| E-ESS1-6.....                              | 21 |
| E-ESS1-6 Academic Language .....           | 23 |
| ESS2 – Earth’s Systems .....               | 24 |
| E-ESS2-1.....                              | 24 |
| E-ESS2-1 Academic Language .....           | 27 |
| E-ESS2-2.....                              | 28 |
| E-ESS2-2 Academic Language .....           | 29 |
| E-ESS2-3.....                              | 31 |
| E-ESS2-3 Academic Language .....           | 34 |
| E-ESS2-4.....                              | 35 |
| E-ESS2-4 Academic Language .....           | 38 |
| E-ESS2-5.....                              | 39 |
| E-ESS2-5 Academic Language .....           | 41 |
| E-ESS2-6.....                              | 43 |

|                                                                                                                                   |    |
|-----------------------------------------------------------------------------------------------------------------------------------|----|
| E-ESS2-6 Academic Language .....                                                                                                  | 44 |
| E-ESS2-7 .....                                                                                                                    | 45 |
| E-ESS2-7 Academic Language .....                                                                                                  | 46 |
| ESS3 – Earth and Human Activity .....                                                                                             | 48 |
| E-ESS3-1 .....                                                                                                                    | 48 |
| E-ESS3-1 Academic Language .....                                                                                                  | 50 |
| E-ESS3-2 .....                                                                                                                    | 51 |
| E-ESS3-2 Academic Language .....                                                                                                  | 53 |
| E-ESS3-3 .....                                                                                                                    | 54 |
| E-ESS3-3 Academic Language .....                                                                                                  | 56 |
| E-ESS3-4 .....                                                                                                                    | 58 |
| E-ESS3-4 Academic Language .....                                                                                                  | 59 |
| E-ESS3-5 .....                                                                                                                    | 61 |
| E-ESS3-5 Academic Language .....                                                                                                  | 62 |
| E-ESS3-6 .....                                                                                                                    | 64 |
| E-ESS3-6 Academic Language .....                                                                                                  | 65 |
| E-ESS3-7 .....                                                                                                                    | 67 |
| E-ESS3-7 Academic Language .....                                                                                                  | 69 |
| References .....                                                                                                                  | 70 |
| Appendix A: Earth and Space Science Grade Vertical Articulation of Disciplinary Core Idea<br>Instructional Focus Statements ..... | 71 |
| ESS1 – Earth's Place in the Universe .....                                                                                        | 72 |
| E-ESS1-1 .....                                                                                                                    | 72 |
| E-ESS1-2 .....                                                                                                                    | 73 |
| E-ESS1-3 .....                                                                                                                    | 75 |
| E-ESS1-4 .....                                                                                                                    | 76 |
| E-ESS1-5 .....                                                                                                                    | 78 |
| E-ESS1-6 .....                                                                                                                    | 79 |
| ESS2 – Earth Systems .....                                                                                                        | 81 |
| E-ESS2-1 .....                                                                                                                    | 81 |
| E-ESS2-2 .....                                                                                                                    | 82 |
| E-ESS2-3 .....                                                                                                                    | 84 |
| E-ESS2-4 .....                                                                                                                    | 88 |

|                                       |     |
|---------------------------------------|-----|
| E-ESS2-5.....                         | 90  |
| E-ESS2-6.....                         | 91  |
| E-ESS2-7.....                         | 92  |
| ESS3 – Earth and Human Activity ..... | 94  |
| E-ESS3-1.....                         | 94  |
| E-ESS3-2.....                         | 96  |
| E-ESS3-3.....                         | 99  |
| E-ESS3-4.....                         | 101 |
| E-ESS3-5.....                         | 104 |
| E-ESS3-6.....                         | 105 |
| E-ESS3-7.....                         | 107 |

## **Purpose and Use**

Science is a way of understanding the physical universe using observation and experimentation to explain natural phenomena. Science also refers to an organized body of knowledge that includes core ideas to the disciplines and common themes that bridge the disciplines. As science educators we must take a 3-dimensional approach in facilitating student learning. By addressing content, science and engineering practices and crosscutting concepts, students can have relevant and evidence-based instruction that can help solve current and future problems.

This document is intended as a guide for discerning and describing features of students and their work who have met the stated Performance Expectation (PE). This document is not intended to be read from cover to cover, but to be used, when needed, to support teacher professional learning and curriculum decisions. This is not intended for student use, and thus is not written in student-friendly language. This is not a curriculum or a means to limit instruction in the classroom. Although each PE states a dedicated Science and Engineering Practice (SEP) and Crosscutting Concept (CCC), students will need to use the whole range of SEPs and CCCs to achieve success by the end of instruction.

Three-dimensional science learning requires discipline specific communication skills. This means that effective science learning occurs when students are expected to speak, listen, read, and write in ways that are appropriate to science. With each Performance Target, there are question/sentence stems and terminology to support student discourse about phenomena to help teachers facilitate the acquisition of science discourse. Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding. The terms and stems in this section are intended to provide a baseline for teachers, neither list is exhaustive nor complete.

In addition to the doing (SEP), thinking (CCC), and learning of science knowledge (Disciplinary Core Ideas) outlined here, students will also require a working knowledge of grade-level appropriate tools and techniques of science. Students should know and recognize how scientists and engineers use these tools and techniques, not just identify them. Students should be able to use these tools to gather data, describe how these tools gather data, and/or interpret data sampled from them. Students will need to understand and apply the conventions of scientific notation when working with extremely large or small quantities of measurement and their calculations.

## Document Updates

### July 2026

- Addition of Appendix A: Earth and Space Science Vertical Articulation of Disciplinary Core Idea Instructional Focus Statements
- Adjusted formatting and grammar.

### July 2025

- All Performance Expectation statements have been reformatted to call out each of the dimensions as follows:
  - Science and Engineering Practice – **bold**
  - Crosscutting Concept – *italicize*
  - Disciplinary Core Idea – regular
- The watermark from previous versions of this resource has been replaced with the wording “For use 2025-2026” on the title page and in the footer. This change was made to improve accessibility of this resource.
- Because scientific notation is no longer an expectation of the math standards, the following statement was added to the purpose and use page to support the teaching and understanding of scientific notation: “Students will need to understand and apply the conventions of scientific notation when working with extremely large or small quantities of measurement and their calculations.”

### June 2024

- Updated watermark to 2024-2025.
- Adjusted formatting and grammar.

## ESS1 – Earth’s Place in the Universe

**E-ESS1-1. Develop a model based on evidence to illustrate that** energy generated by nuclear fusion within the sun (and other stars) *radiates to and influences orbiting planets.*

**Clarification Statement:** Emphasis should be on the energy from nuclear fusion in a star’s core (relative to the star’s mass and age) radiating to nearby planets as seen in the Earth-Sun system. This energy varies in cyclic and non-cyclic ways over the lifespan of the star. Examples of evidence could include observations of other solar systems, surface fluctuations, electromagnetic radiation emissions, atmospheric interactions, solar incidence, and albedo.

**State Assessment Boundary:** Assessment does not include details of the mechanism of nuclear fusion. Assessment does not include details of the atomic and subatomic processes involved with the sun’s nuclear fusion.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                  | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Crosscutting Concepts                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developing and Using Models</b><br><br>Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed world(s).<br><br>Develop a model (including mathematical and computational) to generate data to support explanations, predict phenomena, and analyze systems. | <b>ESS1.A: The Universe and Its Stars</b><br><br>Nuclear fusion within stars releases electromagnetic energy (seen as starlight). Stars go through a sequence of developmental stages over their lifespans-- they are formed; evolve in size, mass, and brightness; and eventually burn out. The Sun is a medium sized star that is about halfway through its predicted life span of approximately 10 billion years. The Sun is just one of more than 200 billion stars in the Milky Way galaxy, and the Milky Way is just one of hundreds of billions of galaxies in the universe.<br><br><b>PS3.D: Energy in Chemical Processes and Everyday Life</b><br><br>Nuclear fusion processes in the center of the sun release the energy that ultimately reaches Earth as radiation. ( <i>secondary</i> ) | <b>Energy and Matter</b><br><br>Energy cannot be created or destroyed—only moves between one place and another place, between objects and/or fields, or between systems. |

Observable features of student performance by the end of the course:

**1. Components of the model**

- a. Students develop/use a model and identify relevant components, including:
  - i. hydrogen as the Sun's (and other stars') fuel,
  - ii. helium and energy (photons) as the products of fusion processes in the Sun (and other stars),
  - iii. solar energy (for example: visible light, ultraviolet light, gamma rays, etc.), and
  - iv. orbiting planets.

**2. Relationships**

- a. Students develop/use a model and describe relationships among the components, including:
  - i. the fusion of hydrogen in the Sun (and other stars) core to produce helium and energy (photons),
  - ii. the release of solar energy from the surface of the Sun, and
  - iii. that released solar energy influences orbiting planets.

**3. Connections**

- a. Students develop/use a model to predict how the relative proportions of hydrogen and helium change as the Sun ages.
- b. Students develop/use a model to describe the scale of energy released by the fusion process as being much larger than the energy released by chemical processes.
- c. Students develop/use a model to describe the influence of solar energy (for example: visible light, ultraviolet light, gamma rays, etc.), on orbiting planets.

## ***E-ESS1-1 Academic Language***

### Question/Sentence Stems

- One evidence we have for the cycling of matter in this system is \_\_\_\_\_.
- The flow/transfer of energy causes \_\_\_\_\_ to occur in the system.
- Energy transfer mechanisms include \_\_\_\_\_.
- When energy leaves the system, the energy \_\_\_\_\_.
- During fusion, the matter in the system is \_\_\_\_\_ and the energy in the system \_\_\_\_\_.
- The (varying) mass of stars exemplifies how matter \_\_\_\_\_ and how energy \_\_\_\_\_ through the system.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                                 |                              |
|---------------------------------|------------------------------|
| • albedo                        | • nuclear fusion             |
| • atmosphere                    | • nucleosynthesis            |
| • atomic nuclei                 | • nucleus                    |
| • convection                    | • photon                     |
| • core                          | • proton                     |
| • electromagnetic radiation     | • radiation                  |
| • energy                        | • solar flare                |
| • fission                       | • solar maximum              |
| • gamma ray                     | • solar minimum              |
| • heavy elements                | • solar storm                |
| • helium                        | • star life cycle            |
| • hydrogen                      | • subatomic particle         |
| • infrared (IR) radiation       | • sunspot                    |
| • Law of Conservation of Energy | • sunspot cycle              |
| • Law of Conservation of Mass   | • ultraviolet (UV) radiation |
| • luminosity                    | • ultraviolet light          |
| • mass                          | • visible light              |
| • neutron                       |                              |

**E-ESS1-2. Construct an explanation of the Big Bang Theory** based on evidence to show that the universe is *changing over time*.

**Clarification Statement:** Emphasis is on astronomical evidence that shows the expansion, cooling, and observed composition of the universe. Examples of supporting data include red shift of light from receding galaxies, cosmic microwave background radiation, and spectra of electromagnetic radiation from stars and interstellar gases that match predictions from models of the Big Bang theory.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Crosscutting Concepts                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Constructing Explanations and Designing Solutions</b></p> <p>Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories.</p> <p>Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.</p> | <p><b>ESS1.A: The Universe and Its Stars</b></p> <p>The Big Bang theory is supported by observations of distant galaxies receding from our own, of the measured composition of stars and non-stellar gases, and of the maps of spectra of the primordial radiation (cosmic microwave background) that still fills the universe. Other than the hydrogen and helium formed at the time of the Big Bang, nuclear fusion within stars produces all atomic nuclei lighter than and including iron, and the process releases electromagnetic energy. Stars' radiation of visible light and other forms of energy can be measured and studied to develop explanations about the formation, age, and composition of the universe.</p> <p>The study of stars' light spectra and brightness is used to identify compositional elements of stars, their movements, and their distances from Earth.</p> <p><b>PS4.B: Electromagnetic Radiation</b></p> <p>Atoms of each element emit and absorb characteristic frequencies of light. These characteristics allow identification of the presence of an element, even in microscopic quantities. (<i>secondary</i>)</p> <p><b>ETS2.A: Interdependence of Science, Engineering, and Technology</b></p> <p>Science and engineering complement each other in the cycle known as research and development (R&amp;D). Many R&amp;D projects may involve scientists, engineers, and others with wide ranges of expertise. (<i>secondary</i>)</p> | <p><b>Stability and Change</b></p> <p>Much of science deals with constructing explanations of how things change and how they remain stable.</p> |

Observable features of student performance by the end of the course:

**1. Articulating the explanations of phenomena**

- a. Students articulate a statement describing/explaining that astronomical evidence from multiple sources is used collectively to support the Big Bang theory and that the universe continues to change over time.

**2. Evidence**

- a. Students identify and describe the evidence to construct the explanation, including:
  - i. the composition (hydrogen, helium, and heavier elements) of stars,
  - ii. the hydrogen-helium ratio of stars and interstellar gases,
  - iii. the redshift of the majority of galaxies and the redshift vs. distance relationship, and
  - iv. the existence of cosmic background radiation.

**3. Reasoning**

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
  - i. Redshifts indicate that an object is moving away from the observer, thus the observed redshift for most galaxies and the redshift vs. distance relationship is evidence that the universe is expanding.
  - ii. The observed cosmic background radiation and the ratio of hydrogen to helium have been shown to be consistent with a universe that was very dense and hot a long time ago and evolved through different stages as it expanded and cooled.
  - iii. An expanding universe must have been smaller in the past and can be extrapolated back in time to a tiny size from which it expanded.

## ***E-ESS1-2 Academic Language***

### Question/Sentence Stems

- The things that stay the same are \_\_\_\_\_.
- The things that change are \_\_\_\_\_.
- The things that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_(event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing this system to be unstable.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                                         |                               |
|-----------------------------------------|-------------------------------|
| • absolute magnitude                    | • Hubble's Law                |
| • absorption                            | • hydrogen                    |
| • apparent magnitude                    | • Law of Conservation of Mass |
| • Big Bang Theory                       | • light-year                  |
| • blueshift                             | • non-stellar gases           |
| • cosmic microwave background radiation | • parallax                    |
| • Doppler effect                        | • recessional velocity        |
| • electromagnetic radiation             | • redshift                    |
| • element                               | • reflection                  |
| • emission                              | • refraction                  |
| • expanding universe                    | • spectra                     |
| • fission                               | • spectroscope                |
| • frequency                             | • spectrum                    |
| • fusion                                | • star                        |
| • galaxy                                | • telescope                   |
| • galaxy cluster                        | • visible light               |
| • helium                                | • wavelength                  |

**E-ESS1-3. Construct an explanation using evidence to explain *the ways elements are produced over the life cycle of a star.***

**Clarification Statement:** Emphasis is on the way nucleosynthesis, and therefore the different elements created, varies as a function of the mass of a star and the stage of its lifetime. Emphasis is on the concept that the matter found in our solar system originated from the deaths of other stars. Examples of evidence include data from stars such as composition, temperature, size, mass, and luminosity.

**State Assessment Boundary:** Assessment does not include details of the atomic and subatomic processes involved with nucleosynthesis.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Crosscutting Concepts                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Constructing Explanations and Designing Solutions</b></p> <p>Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories.</p> <p>Construct and revise an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.</p> | <p><b>ESS1.A: The Universe and Its Stars</b></p> <p>The study of stars' light spectra and brightness is used to identify compositional elements of stars. Stars go through a sequence of developmental stages--they are formed; evolve in size, mass, and brightness; and eventually burn out.</p> <p>Other than the hydrogen and helium formed at the time of the Big Bang, nuclear fusion within stars produces all atomic nuclei lighter than and including iron. Heavier elements are produced when certain massive stars achieve a supernova stage and explode. Material from earlier stars that explode as supernovas is recycled to form younger stars and their planetary systems.</p> | <p><b>Energy and Matter</b></p> <p>Energy and matter cannot be created nor destroyed - only moved between one place and another place between objects and/or fields, or between systems.</p> |

Observable features of student performance by the end of the course:

**1. Articulating the explanations of phenomena**

- a. Students articulate a statement describing/explaining relationships between:
  - i. the life cycle of the stars,
  - ii. the production of elements,
  - iii. the conservation of the number of protons plus neutrons in stars, and
  - iv. that atoms are not conserved in nuclear fusion, but the total number of protons plus neutrons is conserved.

**2. Evidence**

- a. Students identify and describe the evidence to construct the explanation, including:
  - i. helium and a small amount of other light nuclei (up to lithium) were formed from high-energy collisions from protons and neutrons,
  - ii. more massive elements, up to iron, are produced in the cores of stars by a chain of nuclear fusion,
  - iii. supernova explosions of massive stars produce elements more massive than iron,
  - iv. correlation between a star's mass and stage of development and the types of elements it can create during its lifetime, and
  - v. electromagnetic emission and absorption spectra are used to determine a star's composition, motion, temperature, size, mass, luminosity, and distance to Earth.

**3. Reasoning**

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
  - i. Helium and a small amount of other light nuclei (up to lithium) were formed from high-energy collisions from protons and neutrons in the early universe before any stars existed.
  - ii. More massive elements, up to iron, are produced in the cores of stars by a chain of nuclear fusion, which also releases energy.
  - iii. Supernova explosions of massive stars produce elements more massive than iron.
  - iv. There is a correlation between a star's mass and stage of development and the types of elements it can create during its lifetime.
  - v. Electromagnetic emission and absorption spectra are used to determine a star's composition, motion, temperature, size, mass, luminosity, and distance to Earth.

### ***E-ESS1-3 Academic Language***

#### Question/Sentence Stems

- The light from distant stars tells us \_\_\_\_\_.
- The composition of matter in the Universe tells us \_\_\_\_\_.
- In this system, energy impacts matter when \_\_\_\_\_.
- During the Big Bang, the matter in the system is \_\_\_\_\_.
- After the Big Bang, the matter in the system is \_\_\_\_\_.
- Using evidence of cosmic background radiation (energy/heat), we can infer \_\_\_\_\_.
- Using evidence of redshift/blueshift of matter (galaxies) (or spectral lines), we can infer \_\_\_\_\_.

#### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                                 |                       |
|---------------------------------|-----------------------|
| • absolute magnitude            | • luminosity          |
| • absorption lines              | • main sequence       |
| • absorption spectrum           | • mass                |
| • apparent magnitude            | • nebulae             |
| • Big Bang Theory               | • neutrinos           |
| • black hole                    | • neutron             |
| • blue giant                    | • nova                |
| • burnout                       | • nuclear reactions   |
| • electromagnetic radiation     | • nucleons            |
| • elements                      | • nucleosynthesis     |
| • emission lines                | • planetary nebula    |
| • emissions spectrum            | • pressure            |
| • endothermic reaction          | • proton              |
| • exothermic reaction           | • protostar           |
| • fission                       | • red giant           |
| • fusion                        | • stability           |
| • gamma rays                    | • star life cycle     |
| • globular cluster              | • supergiant          |
| • gravity                       | • supernova           |
| • heavy elements                | • supernova remnant   |
| • H-R diagram                   | • surface temperature |
| • Law of Conservation of Energy | • telescope           |
| • Law of Conservation of Mass   | • visible light       |
| • light-year                    | • white dwarf         |

**E-ESS1-4. Use mathematical or computational representations** to *predict* the motion of orbiting objects in the universe due to gravity.

**Clarification Statement:** Emphasis is on predicting orbital motion of naturally occurring or human-made objects using Kepler’s laws and Newton’s law of gravity.

**State Assessment Boundary:** Mathematical representations for the gravitational attraction of bodies and Kepler’s laws of orbital motions should not deal with more than two bodies, nor involve calculus.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Crosscutting Concepts                                                                                                                                                                                                  |
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| <p><b>Using Mathematics and Computational Thinking</b></p> <p>Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Simple computational simulations are created and used based on mathematical models of basic assumptions.</p> <p>Use mathematical, computational, and/or algorithmic representations of phenomena to describe and/or support claims and/or explanations.</p> | <p><b>ESS1.B: Earth and the Solar System</b></p> <p>Kepler’s laws describe common features of the motions of orbiting objects, including their elliptical paths around the sun. Orbits may change due to the gravitational effects from, or collisions with, other objects in the solar system.</p> <p><b>ETS2.A: Interdependence of Science, Engineering, and Technology</b></p> <p>Science and engineering complement each other in the cycle known as research and development (R&amp;D). Many R&amp;D projects may involve scientists, engineers, and others with wide ranges of expertise. (<i>secondary</i>)</p> | <p><b>Scale, Proportion, and Quantity</b></p> <p>Algebraic thinking is used to examine scientific data and predict the effect of a change in one variable on another (e.g., linear growth vs. exponential growth).</p> |

Observable features of student performance by the end of the course:

**1. Representation**

- a. Students develop/use mathematical and/or computational models of orbital motion to identify and describe relevant components, including:
  - i. the trajectories of orbiting bodies (for example: planets, moons, human-made spacecraft, etc.) and
  - ii. a revolving body's eccentricity ( $e = \frac{c}{a}$ ; where  $c$  is the distance between foci of an ellipse and  $a$  is the ellipse's major axis length [Kepler's first law of planetary motion]).

**2. Mathematical and/or computational modeling**

- a. Students develop/use mathematical and/or computational models of orbital motion to model that the square of a revolving body's period is proportional to the cube of its distance to a gravitational center ( $T^2 \propto R^3$ ; where  $T$  is the orbital period and  $R$  is the semi-major axis of the orbit [Kepler's third law of planetary motion]).

**3. Analysis**

- a. Students develop/use and analyze a mathematical and/or computational model of Kepler's second law of planetary motion (an orbiting body sweeps out equal areas in equal time) to predict the relationship between:
  - i. the distance between an orbiting body and its star, and
  - ii. the object's orbital velocity (for example: the closer an orbiting body is to a star, the larger its orbital velocity will be).

### ***E-ESS1-4 Academic Language***

#### Question/Sentence Stems

- To understand the phenomenon of \_\_\_\_\_, I/we can use a scale of \_\_\_\_\_ in my/our model because \_\_\_\_\_.
- Different temporal scales include \_\_\_\_\_. This is important to my model because \_\_\_\_\_.
- Different spatial scales include \_\_\_\_\_. This is important to my model because \_\_\_\_\_.
- Using different temporal and spatial scales allows for \_\_\_\_\_.

#### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- aphelion
- astronomical unit
- eccentricity ( $e = \frac{c}{a}$ )
- ellipse
- foci
- focus
- gravitation
- gravitational constant
- orbit
- perihelion
- period
- revolution
- rotation
- satellite
- semi-major axis
- semi-minor axis
- Kepler's Laws
- axis
- trajectory
- velocity ( $v = \frac{\Delta x}{\Delta t}$ )
- speed ( $s = \frac{d}{t}$ )

**E-ESS1-5. Evaluate evidence of the past and current movements** of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks.

**Clarification Statement:** Emphasis is on the ability of plate tectonics to explain the ages of crustal rocks. Examples include evidence of the ages of oceanic crust increasing with distance from mid-ocean ridges (a result of plate spreading) and the ages of North American continental crust decreasing with distance away from a central ancient core of the continental plate (a result of past plate interactions).

| <b>Science and Engineering Practices</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Crosscutting Concepts</i>                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <p><b>Engaging in Argument from Evidence</b></p> <p>Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current scientific or historical episodes in science.</p> <p>Evaluate evidence behind currently accepted explanations or solutions to determine the merits of arguments.</p> | <p><b>ESS1.C: The History of Planet Earth</b></p> <p>Continental rocks, which can be older than 4 billion years, are generally much older than the rocks of the ocean floor, which are less than 200 million years old. Tectonic processes continually generate new ocean seafloor at ridges and destroy old seafloor at trenches.</p> <p><b>ESS2.B: Plate Tectonics and Large-Scale System Interactions</b></p> <p>The theory of plate tectonics is supported by evidence of ocean floor spreading over time given by tracking magnetic patterns in undersea rocks and coordinating them with changes to Earth’s magnetic axis data.</p> <p>Earth’s history is still being written. Continents are continually being shaped and reshaped by competing constructive and destructive geological processes. North America, for example, has gradually grown in size over the past 4 billion years through a complex set of interactions with other continents, including the addition of many new crustal segments.</p> | <p><b>Patterns</b></p> <p>Empirical evidence is needed to identify patterns.</p> |

Observable features of student performance by the end of the course:

**1. Identifying the explanation and supporting evidence**

- a. Students identify an explanation about how the crustal materials of different ages are arranged on Earth's surface in a pattern that can be attributed to
  - i. plate tectonic activity and
  - ii. formation of new rock from magma rising where plates are moving apart.

**2. Identifying additional evidence relevant to the evaluation**

- a. Students identify and describe additional evidence (for example: data, information, models, etc.) that was not provided but is relevant to the explanation and its evaluation, including:
  - i. measurement of the ratio of parent to daughter atoms produced during radioactive decay to determine the ages of rocks,
  - ii. ages and locations of continental rocks,
  - iii. ages and locations of rocks found on opposite sides of mid-ocean ridges, and
  - iv. type and location of plate boundaries relative to the type, age, and location of crustal rocks.

**3. Evaluating and critiquing**

- a. Students evaluate how the evidence supports or refutes the explanation about the ages of crustal rocks, including:
  - i. continental crust being older than oceanic crust,
  - ii. the oldest continental rocks are located at the center of continents and decrease in age with proximity to mid-ocean ridges, and
  - iii. the oldest oceanic crust is nearest the continents and decrease in age with proximity to mid-ocean ridges.
- b. Students synthesize the evidence to describe the relationship between the motion of continental plates and the ages of crustal rocks, including that:
  - i. At boundaries where plates are moving apart (for example: mid-ocean ridges), material from the interior of Earth must be emerging and forming new rocks with the youngest ages.
  - ii. Regions furthest from plate boundaries (continental centers) have the oldest rocks because new crust is added to the edge of continents at places where plates are coming together (for example: subduction zones).
  - iii. The oldest crustal rocks are found on the continents because oceanic crust is constantly being destroyed at places where plates are coming together (for example: subduction zones).

## ***E-ESS1-5 Academic Language***

### Question/Sentence Stems

- The interactions of plates at boundaries causes \_\_\_\_\_.
- \_\_\_\_\_ are the features associated with a convergent plate boundary.
- \_\_\_\_\_ are the features associated with a divergent plate boundary.
- \_\_\_\_\_ are the features associated with a transform plate boundary.
- \_\_\_\_\_ are the features associated with hotspots.
- The movement of tectonic plates causes \_\_\_\_\_ because \_\_\_\_\_.
- The movement of tectonic plates can be proved by \_\_\_\_\_ because \_\_\_\_\_.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                             |                      |
|-----------------------------|----------------------|
| • asthenosphere             | • magnetometer       |
| • basalt                    | • mantle             |
| • constructive              | • metamorphic        |
| • continental drift theory  | • mid ocean ridge    |
| • continental plate         | • ocean basin        |
| • convection                | • oceanic plate      |
| • convergence               | • paleomagnetism     |
| • convergent plate boundary | • parent atom        |
| • core                      | • plate boundary     |
| • crust                     | • plate tectonics    |
| • daughter atom             | • radioactive decay  |
| • destructive               | • regional hotspot   |
| • divergence                | • reversals          |
| • divergent plate boundary  | • rift valley        |
| • earthquake pattern        | • rock cycle         |
| • geologic process          | • satellite          |
| • igneous                   | • seafloor spreading |
| • lava                      | • sedimentary        |
| • lithosphere               | • sonar              |
| • magma                     | • subduction         |
| • magnetic axis             | • subduction zone    |
| • magnetic field            | • time scale         |
| • magnetic polarity         | • trench             |
| • magnetic strips           | • volcanic           |

**E-ESS1-6. Apply scientific reasoning and evidence from** ancient Earth materials, meteorites, and other planetary surfaces *to construct an account of Earth’s formation* and early history.

**Clarification Statement:** Emphasis is on using available evidence within the solar system to reconstruct the early history of Earth, which formed simultaneously along with the rest of the solar system. Examples of evidence include the absolute ages of ancient materials (obtained by radiometric dating of meteorites, moon rocks, and Earth’s oldest minerals), the sizes and compositions of solar system objects, and the impact cratering record of planetary surfaces.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Crosscutting Concepts                                                                                                                           |
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| <p><b>Constructing Explanations and Designing Solutions</b></p> <p>Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories.</p> <p>Apply scientific reasoning to link evidence to the claims to assess the extent to which the reasoning and data support the explanation or conclusion.</p> | <p><b>ESS1.B: The Earth and the Solar System</b></p> <p>The solar system consists of the Sun and a collection of objects of varying sizes and conditions. This system appears to have formed from a disk of dust and gas, drawn together by gravity approximately 4.6 billion years ago.</p> <p><b>ESS1.C: The History of Planet Earth</b></p> <p>Radioactive decay lifetimes and isotopic content in rocks provide a way of dating rock formations and thereby fixing the scale of geological time.</p> <p>Although active geologic processes, such as plate tectonics and erosion, have destroyed or altered most of the very early rock record on Earth, other objects in the solar system, such as lunar rocks, asteroids, and meteorites, have changed little over billions of years.</p> <p>Studying these objects can provide information about Earth’s formation and early history. Study of other planets and their moons, many of which exhibit features such as volcanism and meteor impacts similar to those found on Earth, also help illuminate aspects of Earth’s history and changes.</p> <p><b>PS1.C: Nuclear Processes</b></p> <p>Spontaneous radioactive decays follow a characteristic exponential decay law. Nuclear lifetimes allow radiometric dating to be used to determine the ages of rocks and other materials thereby fixing the scale of geological time.</p> <p>(secondary)</p> | <p><b>Stability and Change</b></p> <p>Much of science deals with constructing explanations of how things change and how they remain stable.</p> |

Observable features of student performance by the end of the course:

**1. Articulating the explanation of phenomena**

- a. Students articulate a statement describing/explaining Earth's formation and early history, including:
  - i. Earth formed along with the rest of the solar system 4.6 billion years ago.
  - ii. Early Earth was bombarded just like other objects in the solar system were bombarded.
  - iii. Erosion and plate tectonics on Earth have destroyed much of the evidence of this bombardment, thus the relative scarcity of impact craters on Earth.

**2. Evidence**

- a. Students identify and describe the evidence to construct the explanation, including:
  - i. age and composition of Earth's oldest rocks, lunar rocks, and meteorites as determined by radiometric dating,
  - ii. composition of solar system objects,
  - iii. observations of the size and distribution of impact craters on the surface of Earth and other solar system objects (for example: the Moon, Mercury, Mars, etc.), and
  - iv. activity of plate tectonic processes, such as volcanism, and surface processes, such as erosion, on Earth.

**3. Reasoning**

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
  - i. The radiometric ages of lunar rocks, meteorites, and the oldest Earth rocks provide evidence for an origin of the solar system 4.6 billion years ago, with the creation of a solid Earth crust about 4.4 billion years ago.
  - ii. Other planetary surfaces and their patterns of impact cratering can be used to infer that Earth had many impact craters early in its history.
  - iii. The relative lack of impact craters and the age of most rocks on Earth compared to other bodies in the solar system can be attributed to processes like volcanism, plate tectonics, and erosion that have reshaped Earth's surface, and that is why most of Earth's rocks are much younger than Earth itself.

## ***E-ESS1-6 Academic Language***

### Question/Sentence Stems

- The data shows that \_\_\_\_\_ changes over time.
- \_\_\_\_\_ supports the idea that \_\_\_\_\_ has changed over geologic time.
- The components that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_ (event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing/caused this system to be unstable.
- Evidence for Earth's formation includes \_\_\_\_\_.
- Evidence for changes to Earth's systems since its formation includes \_\_\_\_\_.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                          |                         |
|--------------------------|-------------------------|
| • absolute age           | • mass extinction       |
| • accretion              | • meteor                |
| • ancient core           | • meteorite             |
| • asteroid               | • mineral               |
| • asthenosphere          | • nuclear               |
| • bedrock                | • ocean ridge           |
| • billion                | • ocean trench          |
| • Cambrian               | • oceanic crust         |
| • continental crust      | • particulates          |
| • convection current     | • planetary surfaces    |
| • crater                 | • planetesimals         |
| • cycle                  | • plate tectonics       |
| • erosion                | • Precambrian           |
| • exponential            | • primordial atmosphere |
| • geologic time          | • radioactive decay     |
| • gravitational collapse | • radiometric dating    |
| • gravity                | • relative age          |
| • half-life              | • rock record           |
| • inner core             | • seafloor spreading    |
| • isotope                | • sedimentation         |
| • lithosphere            | • solar nebula          |
| • mantle                 | • volcanism             |

## ESS2 – Earth’s Systems

**E-ESS2-1. Use evidence to argue how** Earth’s internal and external processes operate to *form and modify* continental and ocean-floor features throughout Earth’s history.

**Clarification Statement:** Emphasis is on the core idea that convection leads to the creation and destruction of surface features. Plate movements and many crustal features and events are a result of this phenomenon, but there are other surface processes, which shape Earth’s surface as well. The appearance of land features (such as mountains, valleys, coastlines, and plateaus) and sea-floor features (such as trenches, ridges, and seamounts) are a result of both constructive forces (such as volcanism, tectonic uplift, and orogeny) and destructive mechanisms (such as weathering, mass wasting, and coastal erosion). Examples include seafloor spreading at ridges (evidenced by paleomagnetic data and radiometric dating of rocks), subduction at trenches (evidenced by seismic data and volcanoes), and weathering and erosion among mountains (evidenced by weathering, erosion, and deposition patterns of streams).

**State Assessment Boundary:** Assessment does not include memorization of the details of the formation of specific geographic features of Earth’s surface.

| <b>Science and Engineering Practices</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <i>Crosscutting Concepts</i>                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Engaging in Argument from Evidence</b></p> <p>Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current scientific or historical episodes in science.</p> <p>Construct an oral and written argument or counter-arguments based on data and evidence.</p> | <p><b>ESS2.A: Earth Materials and Systems</b></p> <p>Earth’s systems, being dynamic and interacting, cause feedback effects that can increase or decrease the original changes. A deep knowledge of how feedback works within and among Earth’s systems is still lacking, thus limiting scientists’ ability to predict some changes and their impacts.</p> <p>The top part of the mantle, along with the crust, forms structures known as tectonic plates.</p> <p>These changes can occur on a variety of time scales from sudden (e.g., volcanic ash clouds) to intermediate (ice ages) to long-term tectonic cycles.</p> <p><b>ESS2.B: Plate Tectonics and Large-Scale System Interactions</b></p> <p>The radioactive decay of unstable isotopes continually generates new energy within Earth’s crust and mantle providing the primary source of the heat that drives mantle convection. Plate tectonics can be viewed as the surface expression of mantle convection.</p> <p>The plates move across Earth’s surface, carrying the continents, creating and destroying ocean basins, producing earthquakes and volcanoes, and forming mountain ranges and plateaus.</p> <p>Most continental and ocean floor features are the result of geological activity and earthquakes along plate boundaries. The exact patterns depend on whether the plates are being pushed together to create mountains or deep ocean trenches, being pulled apart to form new ocean floor at mid-ocean ridges, or sliding past each other along surface faults.</p> | <p><b>Stability and Change</b></p> <p>Much of science deals with constructing explanations of how things change and how they remain stable.</p> |

Observable features of student performance by the end of the course:

**1. Developing a claim**

- a. Students make a claim that Earth's internal and external processes form and modify different continental and ocean-floor features throughout Earth's history.

**2. Identifying scientific evidence**

- a. Students identify and describe the evidence that supports the claim, including:
  - i. descriptions and locations of specific continental and ocean-floor features,
  - ii. internal processes (for example: volcanism, uplift, etc.) and surface processes (for example: weathering, erosion, etc.), and
  - iii. temporal data about the relative times over which processes act to produce continental and/or ocean-floor features.

**3. Evaluating and critiquing evidence**

- a. Students identify strengths and weaknesses of the evidence to support the claim, including:
  - i. types of sources,
  - ii. sufficiency, including validity and reliability, of the evidence to make and defend the claim, and/or
  - iii. any alternative interpretations of the evidence and why the evidence supports or does not support the student's claim, as opposed to any other claims.

**4. Reasoning and synthesis**

- a. Students use the following chain of reasoning to connect the evidence:
  - i. Specific internal processes (for example: volcanism, mountain building, uplift, etc.) are identified as causal agents in building up Earth's surface over time.
  - ii. Specific surface processes (for example: weathering, erosion, etc.) are identified as causal agents in wearing down Earth's surface over time.
  - iii. Interactions and feedback between processes are identified (for example: mountain-building changes weather patterns that then change the rate of erosion of mountains).
  - iv. The rate at which a feature changes is related to the time scale on which the processes operate. Features can form or change slowly due to processes that act on long time scales (for example: continental positions due to plate drift) or rapidly due to processes that act on short time scales (for example: volcanic eruptions).

## ***E-ESS2-1 Academic Language***

### Question/Sentence Stems

- The things that stay the same are \_\_\_\_\_.
- The things that change are \_\_\_\_\_.
- The things that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_(event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing this system to be unstable.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                             |                        |                            |
|-----------------------------|------------------------|----------------------------|
| • coastal erosion           | • fault                | • radioactive decay        |
| • constructive forces       | • feedback effect      | • radiometric dating       |
| • continental boundary      | • geochemical reaction | • recrystallization        |
| • continental crust         | • geologic activity    | • rock composition         |
| • convection                | • heat                 | • seafloor spreading       |
| • convergent plate boundary | • inner core           | • seamount                 |
| • crustal deformation       | • irreversible         | • seismic waves            |
| • crustal plate movement    | • isotope              | • slab drag                |
| • density                   | • mantle               | • sonar                    |
| • deposition                | • mass wasting         | • stream                   |
| • destructive forces        | • mountain             | • subduction               |
| • divergent plate boundary  | • nuclear              | • tectonic uplift          |
| • Earth's magnetic field    | • ocean basin          | • thermal convection       |
| • earthquake                | • ocean ridge          | • time scale               |
| • electromagnetic radiation | • ocean trench         | • transform plate boundary |
| • energy                    | • oceanic crust        | • trench                   |
| • erosion                   | • orogeny              | • unstable atomic nucleus  |
|                             | • outer core           | • uplift                   |
|                             | • paleomagnetic data   | • valley                   |
|                             | • plate tectonics      | • volcanism                |
|                             | • plateau              | • weathering               |
|                             | • pressure             |                            |

**E-ESS2-2. Analyze data to make the claim that one change to Earth’s surface *can create* feedbacks** that cause changes to other Earth systems.

**Clarification Statement:** Examples should include climate feedback, such as how an increase in greenhouse gases causes a rise in global temperatures that melt glacial ice, which reduces the amount of sunlight reflected from Earth’s surface, increasing surface temperatures, and further reducing the amount of ice. Examples could also be taken from other system interactions, such as how the loss of ground vegetation causes an increase in water runoff and soil erosion, how dammed rivers increase groundwater recharge, decrease sediment transport, and increase coastal erosion.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                        | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Crosscutting Concepts                                                                                                                                                                  |
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| <p><b>Analyzing and Interpreting Data</b></p> <p>Analyzing data in 9-12 builds on K-8 experiences and progresses to introducing more detailed statistical analysis, the comparison of data sets for consistency, and the use of models to generate and analyze data.</p> <p>Analyze data using tools, technologies, and/or models (e.g., computational, mathematical) in order to make valid and reliable scientific claims or determine an optimal design solution.</p> | <p><b>ESS2.A: Earth Materials and Systems</b></p> <p>Earth’s systems, being dynamic and interacting, cause feedback effects that can increase or decrease the original changes. Transfers of energy and the movements of matter can cause chemical and physical changes among Earth’s materials and living organisms.</p> <p><b>ESS2.D: Weather and Climate</b></p> <p>The foundation for Earth’s global climate systems is the electromagnetic radiation from the sun, as well as its reflection, absorption, storage, and redistribution among the atmosphere, ocean, and land systems, and this energy’s re-radiation into space.</p> <p><b>ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World</b></p> <p>New technologies can have deep impacts on society and the environment, including some that were not anticipated. Analysis of costs and benefits is a critical aspect of decisions about technology. (<i>secondary</i>)</p> | <p><b>Stability and Change</b></p> <p>Change and rates of change can be quantified and modeled over very short or very long periods of time. Some system changes are irreversible.</p> |

Observable features of student performance by the end of the course:

**1. Organizing data**

- a. Students organize data that represent measurements of changes in hydrosphere, cryosphere, atmosphere, biosphere, or geosphere in response to a change in Earth's surface.

**2. Identifying relationships**

- a. Students analyze data to determine the relationship between:
  - i. changes in one system and changes in another (or within the same) Earth system and
  - ii. possible feedbacks, including one example of feedback to the climate.
- b. Students analyze data to identify effects of human activity and specific technologies on Earth's systems.

**3. Interpreting data**

- a. Students use the analyzed data to describe a mechanism for the feedbacks between two of Earth's systems and whether the feedback is positive or negative, increasing (destabilizing) or decreasing (stabilizing) the original changes.
- b. Students use the analyzed data to describe an unanticipated or unintended effect of a selected technology on Earth's systems.

***E-ESS2-2 Academic Language***

Question/Sentence Stems

- The things that stay the same are \_\_\_\_\_.
- The things that change are \_\_\_\_\_.
- The things that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_(event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing this system to be unstable.
- \_\_\_\_\_ is an example of positive feedback.
- \_\_\_\_\_ is an example of negative feedback.
- \_\_\_\_\_ (increased/decreased) albedo will result in \_\_\_\_\_ (increased/decreased) temperature, which in turn will result in \_\_\_\_\_ (increased/decreased) albedo. This is an example of a \_\_\_\_\_ (positive/negative) feedback.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- absorption
- acidification
- albedo
- anthropogenic changes
- atmosphere
- atmospheric circulation
- atmospheric composition
- biosphere
- carbon cycle
- chemical change
- climate change
- coastal erosion
- convection cycle
- cryosphere
- destabilizing
- electromagnetic radiation
- energy
- feedback effect
- geoscience
- geosphere
- glacier
- greenhouse gas
- ground vegetation
- groundwater recharge
- hydrosphere
- mean surface temperature
- methane
- negative feedback loop
- ocean circulation
- physical change
- positive feedback loop
- redistribution
- reflection
- re-radiation
- runoff
- sea level
- sediment transport
- soil erosion
- solar energy
- stabilizing
- technology

**E-ESS2-3. Develop a model based on evidence of Earth’s interior that describes *cycling of matter*** through convection processes.

**Clarification Statement:** Emphasis is on both a one-dimensional model of Earth, with radial layers determined by density, and a three- dimensional model, which is controlled by mantle convection and the resulting plate tectonics. Examples of evidence include maps of Earth’s three-dimensional structure obtained from seismic waves, records of the rate of change of Earth’s magnetic field (as constraints on convection in the outer core), volcanoes, and identification of the composition of Earth’s layers from high-pressure laboratory experiments.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                    | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Crosscutting Concepts                                                                                                                                                                                                                                              |
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| <p><b>Developing and Using Models</b></p> <p>Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed world(s).</p> <p>Develop a model based on evidence to illustrate and/or predict the relationships between systems or between components.</p> | <p><b>ESS2.A: Earth Materials and Systems</b></p> <p>Evidence from deep probes and seismic waves, reconstructions of historical changes in Earth’s surface and its magnetic field, and an understanding of physical and chemical processes lead to a model of Earth with a hot but solid inner core, a liquid outer core, a solid mantle, and crust. All of Earth’s processes are the result of energy flowing and matter cycling within and among Earth systems. Motions of the mantle and its plates occur primarily through thermal convection, which involves the cycling of matter due to the outward flow of energy from Earth’s interior and the gravitational movement of denser materials toward the interior.</p> <p><b>ESS2.B: Plate Tectonics and Large-Scale System Interactions</b></p> <p>The radioactive decay of unstable isotopes continually generates new energy within Earth’s crust and mantle providing the primary source of the heat that drives mantle convection. Plate tectonics can be viewed as the surface expression of mantle convection. The top part of the mantle, along with the crust, make up the moving tectonic plates of the lithosphere. Tectonic plates ride above giant convection cells that bring matter from the hot inner mantle up to the cool surface. The plates move across Earth’s surface, carrying the continents, creating and destroying ocean basins, producing earthquakes and volcanoes, and forming mountain ranges and plateaus.</p> <p><b>PS4.A: Wave Properties</b></p> <p>Geologists use seismic waves and their reflection at interfaces between layers to probe structures deep in the planet. (<i>secondary</i>)</p> <p><b>ETS2.A: Interdependence of Science, Engineering, and Technology</b></p> <p>Science and engineering complement each other in the cycle known as research and development (R&amp;D). Many R&amp;D projects may involve scientists, engineers, and others with wide ranges of expertise. (<i>secondary</i>)</p> | <p><b>Energy and Matter</b></p> <p>Energy and matter cannot be created nor destroyed - only moved between one place and another place between objects and/or fields, or between systems. The total amount of energy and matter in closed systems is conserved.</p> |

Observable features of student performance by the end of the course:

### **1. Components of the model**

- a. Students develop/use a model (for example: graphical, verbal, mathematical, etc.) and identify the relevant components based on seismic and magnetic evidence (for example: pattern of the geothermal gradient or heat flow measurements) from Earth's interior, including:
  - i. Earth's interior in cross-section and radial layers (crust, mantle, liquid outer core, solid inner core) determined by density,
  - ii. plate activity in the outer part of the geosphere,
  - iii. radioactive decay and residual thermal energy from the formation of Earth as an energy source,
  - iv. heat loss at the surface of Earth, and
  - v. convection processes that cause hot matter to rise (move away from the center) and cool matter to fall (move toward the center).

### **2. Relationships**

- a. Students develop/use a model to describe the relationships between components, including:
  - i. Energy released by radioactive decay in Earth's crust and mantle and residual thermal energy from Earth's formation provide energy that drives the flow of matter in the mantle.
  - ii. Thermal energy is released at the surface of the Earth as new crust is formed and cooled.
  - iii. The flow of matter by convection in the solid mantle and the sinking of cold, dense crust back into the mantle exert forces on crustal plates that then move, producing tectonic activity.
  - iv. The flow of matter by convection in the liquid outer core generates the Earth's magnetic field.
  - v. Matter is cycled between the crust and the mantle at plate boundaries. Where plates are pushed together, cold crustal material sinks back into the mantle, and where plates are pulled apart, mantle materials can be integrated into the crust, forming new rock.

### **3. Connections**

- a. Students develop/use a model to describe the cycling of matter by thermal convection in Earth's interior, including:
  - i. the flow of matter in the mantle that causes crustal plates to move,
  - ii. the flow of matter in the liquid outer core that generates the Earth's magnetic field, including evidence of polar reversals,
  - iii. the radial layers determined by density in Earth's interior, and
  - iv. the addition of thermal energy released by radioactive decay in Earth's crust and mantle.

### ***E-ESS2-3 Academic Language***

#### Question/Sentence Stems

- In Earth's interior, matter is cycled by \_\_\_\_\_.
- The energy for \_\_\_\_\_ is from \_\_\_\_\_.
- The flow of energy causes \_\_\_\_\_ to occur in the system.
- The energy is entering the system by \_\_\_\_\_.
- In the system, the cycling of matter \_\_\_\_\_.
- The flow of energy causes \_\_\_\_\_ to occur in the system as seen with \_\_\_\_\_ as evidence from the investigation.
- The flow of energy between \_\_\_\_\_ and \_\_\_\_\_ drives the changes to the system as seen by \_\_\_\_\_ because \_\_\_\_\_.
- My/Our diagram shows the flow energy in \_\_\_\_\_ as seen by \_\_\_\_\_ because \_\_\_\_\_.
- The effect of changing temperature on density is \_\_\_\_\_.
- The pattern of density of Earth's interior layers is \_\_\_\_\_.

#### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                                   |                    |                           |
|-----------------------------------|--------------------|---------------------------|
| • chemical process                | • geosphere        | • pressure                |
| • composition                     | • heat loss        | • radial layers           |
| • continental crust               | • inner core       | • radioactive decay       |
| • convection                      | • interface        | • reflection              |
| • convection cell                 | • isotope          | • residual thermal energy |
| • crustal movement                | • lithosphere      | • seismic wave            |
| • cycling of matter               | • magnetic field   | • slab drag               |
| • density                         | • mantle           | • tectonic uplift         |
| • Earth's elements                | • molten rock      | • thermal convection      |
| • Earth's internal energy sources | • mountain range   | • time scale              |
| • earthquake                      | • ocean basin      | • unstable atomic nucleus |
| • geochemical cycle               | • oceanic crust    | • volcano                 |
| • geochemical reaction            | • outer core       |                           |
| • geoscience                      | • physical process |                           |
|                                   | • plate tectonics  |                           |
|                                   | • plateau          |                           |
|                                   | • polar reversal   |                           |

**E-ESS2-4. Use a model to describe how** *causes* of short and long-term variations in the flow of energy into and out of Earth's systems *result in* changes to climate.

**Clarification Statement:** Emphasis is on the relationships between components that affect the input, output, storage, and redistribution of energy on Earth. Emphasis is on specific cause-and-effect relationships between the factors that affect energy flow (into and out of Earth's systems) and their effects on climate over different timescales.

**State Assessment Boundary:** Assessment is limited to one example of a climate change and its associated impacts. Assessment of the results of changes in climate is limited to direct changes in climate such as surface temperatures and precipitation patterns.

| <b>Science and Engineering Practices</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Disciplinary Core Ideas</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Crosscutting Concepts</b>                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Developing and Using Models</b></p> <p>Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed world(s).</p> <p>Use a model based on evidence to illustrate and/or predict the relationships between systems or between components.</p> | <p><b>ESS1.B: Earth and the Solar System</b></p> <p>Cyclical changes in the shape of Earth’s orbit around the sun, together with changes in the orientation of the planet’s axis of rotation, both occurring over tens to hundreds of thousands of years, have altered the intensity and distribution of sunlight falling on Earth. These phenomena cause cycles of ice ages and other gradual climate changes. (<i>secondary</i>)</p> <p><b>ESS2.D: Weather and Climate</b></p> <p>The foundation for Earth’s global climate systems is the electromagnetic radiation from the Sun, as well as its reflection, absorption, storage, and redistribution among the atmosphere, ocean, and land systems, and this energy’s re-radiation into space.</p> <p>Climate changes, which are defined as significant and persistent changes in an area’s average or extreme weather conditions can occur if any of Earth’s systems change. Scientists can infer these changes from geological evidence. Some climate changes in Earth’s history were rapid shifts (caused by natural events, such as volcanic eruptions and meteoric impacts, which suddenly put a large amount of particulate matter into the atmosphere or by abrupt changes in ocean currents, or variations in solar output). Other climate changes were gradual and longer term--due, for example, to solar output variations, or atmospheric changes due to the rise of plants and other life forms that modified the atmosphere via photosynthesis. The timescale of these changes varies from a few to millions of years.</p> <p>Cumulative increases in the atmospheric concentrations of carbon dioxide and other greenhouse gases, whether arising from natural sources or human industrial activity, increase the capacity of Earth to retain energy. Changes in the atmosphere due to human activity have increased carbon dioxide concentrations and thus affect climate.</p> | <p><b>Cause and Effect</b></p> <p>Cause-and-effect relationships can be suggested and predicted for complex natural and human designed systems by examining what is known about smaller scale mechanisms within the system.</p> |

Observable features of student performance by the end of the course:

**1. Components of the model**

- a. Students develop/use a model (conceptual, graphical, physical, etc.) and identify the relevant components, including at least one factor that affects the input of energy, at least one factor that affects the output of energy, and at least one factor that affects the storage and distribution of energy. Factors could include:
  - i. changes in Earth's orbit and the orientation of its axis,
  - ii. changes in the Sun's energy output,
  - iii. configuration of continents resulting from tectonic activity,
  - iv. ocean circulation,
  - v. atmospheric conditions (including water vapor and CO<sub>2</sub>),
  - vi. atmospheric circulation,
  - vii. volcanic activity,
  - viii. glaciation,
  - ix. changes in extent or type of vegetation cover, and
  - x. human activities.
- b. Students describe the relevant different time scales on which the factors operate.

**2. Relationships**

- a. Students develop/use a model to describe the relationships between components, including:
  - i. those that affect input of energy,
  - ii. those that affect output of energy, and
  - iii. those that affect the storage and redistribution of energy.

**3. Connections**

- a. Students develop/use a model to describe the relationship between energy flow in Earth's systems and changes in climate, including:
  - i. specific cause and effect relationships between the factors and the effect on energy flow into and out of Earth's systems and
  - ii. the net effect of all the competing factors in changing the climate.

## ***E-ESS2-4 Academic Language***

### Question/Sentence Stems

- \_\_\_\_\_ happens to matter as it moves within the system.
- In this system, energy is entering by \_\_\_\_\_, doing \_\_\_\_\_ in the system, and leaving the system by \_\_\_\_\_.
- When you put energy into this system \_\_\_\_\_.
- The flow of energy causes \_\_\_\_\_ to occur in the system.
- The energy is entering the system by \_\_\_\_\_.
- In the system, the cycling of matter \_\_\_\_\_.
- The matter in the system enters from \_\_\_\_\_.
- The flow of energy causes \_\_\_\_\_ to occur in the system as seen with \_\_\_\_\_ as evidence from the investigation.
- The flow of energy between \_\_\_\_\_ and \_\_\_\_\_ drives the changes to the system as seen by \_\_\_\_\_ because \_\_\_\_\_.
- \_\_\_\_\_ shows that \_\_\_\_\_ is the evidence that energy is being conserved in this system because \_\_\_\_\_.
- The uneven heating of Earth by the Sun results in \_\_\_\_\_.
- When a surface \_\_\_\_\_ (reflects/absorbs) energy, its temperature will \_\_\_\_\_ (increase/decrease).

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                           |                             |                       |                     |
|---------------------------|-----------------------------|-----------------------|---------------------|
| • absorption              | • cyclical changes          | • hydrosphere         | • reflection        |
| • acidification           | • electromagnetic radiation | • interdependence     | • re-radiation      |
| • albedo                  | • energy flow               | • land system         | • reservoir/sink    |
| • atmosphere              | • energy input              | • meteoric impact     | • respiration       |
| • atmospheric circulation | • energy output             | • nitrogen cycle      | • rotation          |
| • atmospheric composition | • energy storage            | • ocean current       | • sea level         |
| • axis                    | • entropy                   | • ocean system        | • solar flare       |
| • biosphere               | • extreme weather           | • oceanic circulation | • solar output      |
| • carbon cycle            | • fossil fuel               | • orbit               | • solar radiation   |
| • carbon dioxide          | • geosphere                 | • oxygen              | • tectonic activity |
| • climatic pattern        | • glacier                   | • particulate matter  | • vegetation cover  |
| • combustion              | • greenhouse gas            | • photosynthesis      | • volcanic eruption |
|                           | • human activity            | • precipitation       |                     |

**E-ESS2-5. Investigate the ways that** water (given its unique physical and chemical properties) *impacts* various Earth systems.

**Clarification Statement:** Emphasis should be on water’s ability to absorb/store and release energy, transmit sunlight, expand when freezing, and dissolve/transport materials. Examples of system interactions could include the hydrogeologic system (weathering, erosion, deposition, soil formation, groundwater formation, and the rock cycle), energy transfer system (weather and climate), and ecosystems (coral reefs and hydrothermal vents).

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                             | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Crosscutting Concepts                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Planning and Carrying out Investigations</b></p> <p>Planning and carrying out investigations in 9- 12 builds on K-8 experiences and progresses to include investigations that provide evidence for and test conceptual, mathematical, physical, and empirical models.</p> <p>Plan and conduct an investigation individually and collaboratively to produce data to serve as the basis for evidence.</p> | <p><b>ESS2.C: The Roles of Water in Earth’s Surface Processes</b></p> <p>The abundance of liquid water on Earth’s surface and its unique combination of physical and chemical properties are central to the planet’s dynamics. These properties include water’s exceptional capacity to absorb, store, and release large amounts of energy as it changes state; transmit sunlight; expand upon freezing; dissolve and transport materials; and lower the viscosities and melting points of the material when mixed with fluid rocks within the mantle. Each of these properties plays a role in how water affects other Earth systems (e.g., ice expansion contributes to rock erosion, or ocean thermal capacity contributes to moderating temperature variations).</p> | <p><b>Cause and Effect</b></p> <p>Cause-and-effect relationships can be suggested and predicted for complex natural and human designed systems by examining what is known about smaller scale mechanisms within the system. Systems can be designed to cause a desired effect.</p> |

Observable features of student performance by the end of the course:

**1. Identifying the phenomenon under investigation**

- a. Students identify the phenomenon under investigation involving the properties of water and its effect on Earth materials and surface processes.

## **2. Identifying the evidence to address the purpose of the investigation**

- a. Students describe the data that will be collected and the evidence to be derived, including:
  - i. The properties of water, including:
    - 1. heat capacity,
    - 2. density in solid, and liquid states, and
    - 3. polar nature due to its molecular structure.
  - ii. The effect of water properties on energy transfer that causes the patterns of temperature, movement of air, and movement and availability of water at Earth's surface.
  - iii. The mechanical effects of water on Earth materials that can be used to infer the effect of water on Earth's surface processes, for example:
    - 1. stream transportation and deposition using a stream table, which can be used to infer the ability of water to transport and deposit materials,
    - 2. erosion using variations in soil moisture content, which can be used to infer the ability of water to prevent or facilitate movement materials, and/or
    - 3. expansion of water as it freezes, which can be used to infer the ability of water to break rocks into smaller pieces.
  - iv. The chemical effects of water on Earth materials that can be used to infer the effect of water on Earth's surface processes, for example:
    - 1. solubility of different materials in water, which can be used to infer chemical weathering and recrystallization,
    - 2. reaction of iron to rust in water, which can be used to infer the role of water in chemical weathering,
    - 3. data illustrating that water lowers the melting temperature of most solids, which can be used to infer melt generation, and/or
    - 4. data illustrating that water decreases the viscosity of melted rock, affecting the movement of magma and volcanic eruptions.

## **3. Planning the investigation**

- a. Students determine and describe the experimental design, including:
  - i. identifying variables and controls,
  - ii. the experimental procedure,
  - iii. the number of trials,
  - iv. the necessary equipment, materials, and techniques, and/or
  - v. how the data will be collected.

#### **4. Refining the experimental design**

- a. Students evaluate the investigation, including:
  - i. assessing the accuracy and precision of the data collected,
  - ii. limitations of the investigation, and/or
  - iii. strength of the data
- b. Students refine the investigation to provide more accurate, precise, and useful data.

#### ***E-ESS2-5 Academic Language***

##### **Question/Sentence Stems**

- By looking at patterns in the data, I/we determined that \_\_\_\_\_ caused \_\_\_\_\_.
- \_\_\_\_\_ caused the patterns I am observing. I know this because \_\_\_\_\_.
- If \_\_\_\_\_ happens, I/we predict that \_\_\_\_\_ will occur.
- Even though I/we cannot see \_\_\_\_\_, it explains why \_\_\_\_\_ is happening.
- When I/we change \_\_\_\_\_ in the system, \_\_\_\_\_ is affected.
- The evidence \_\_\_\_\_ presented in the scenario supports the claim that \_\_\_\_\_ causes \_\_\_\_\_.
- In order to conclude that \_\_\_\_\_ caused \_\_\_\_\_, the following evidence is needed \_\_\_\_\_.
- Water causes changes to Earth's surface by \_\_\_\_\_.
- The property of water \_\_\_\_\_ will result in \_\_\_\_\_ changes to Earth's surface.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- absorption
- adhesion
- atmosphere
- biosphere
- chemical property
- chemical weathering
- climate
- cohesion
- coral reef
- density
- deposition
- dissolve
- ecosystem
- energy release
- energy transfer
- erosion
- fluid rocks
- freezing point
- frost wedging
- geosphere
- groundwater formation
- heat capacity
- hydrogen (H) bonding
- hydrogeologic system
- hydrologic cycle
- hydrosphere
- hydrothermal vent
- ice expansion
- iron
- lava
- magma
- material transport
- mechanical erosion
- melting point
- molecular structure
- phase change
- physical property
- polarity
- pressure
- recrystallization
- rock cycle
- rust
- sediment
- soil formation
- soil moisture content
- solar energy
- solubility
- stream deposition
- stream table
- stream transportation
- surface tension
- thermal
- viscosity
- volcanic eruption
- water property
- weather

**E-ESS2-6. Develop a quantitative model to describe the cycling of carbon through the hydrosphere, atmosphere, geosphere, and biosphere.**

**Clarification Statement:** Emphasis is on modeling biogeochemical cycles that include the cycling of carbon through the ocean, atmosphere, soil, and biosphere (photosynthesis, chemosynthesis, cellular respiration), providing the foundation for living organisms.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                    | Disciplinary Core Ideas                                                                                                                                                                                                                                                                            | Crosscutting Concepts                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <p><b>Developing and Using Models</b></p> <p>Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed world(s).</p> <p>Develop a model based on evidence to illustrate and/or predict the relationships between systems or between components.</p> | <p><b>ESS2.D: Weather and Climate</b></p> <p>Gradual atmospheric changes were due to plants and other organisms that captured carbon dioxide and released oxygen.</p> <p>Changes in the atmosphere due to human activity have increased carbon dioxide concentrations and thus affect climate.</p> | <p><b>Energy and Matter</b></p> <p>The total amount of energy and matter in closed systems is conserved.</p> |

Observable features of student performance by the end of the course:

**1. Components of the model**

- a. Students develop/use a model (conceptual, graphical, physical, etc.) and identify the relevant components, including:
  - i. relevant concentrations of carbon in the hydrosphere, atmosphere, geosphere, and biosphere and
  - ii. represent carbon cycling from one sphere to another.

**2. Relationships**

- a. Students develop/use a model to describe the relationships between components, including:
  - i. biogeochemical cycles that occur as carbon flows from one sphere to another,
  - ii. relative amount of and rate at which carbon is transferred between spheres,
  - iii. capture of carbon by plants, and
  - iv. increase in carbon dioxide concentration in the atmosphere due to human activity and the effect on climate.

### 3. Connections

- a. Students develop/use a model to explicitly identify the conservation of matter as carbon cycles through various components of Earth's systems.
- b. Students identify limitations of the model in accounting for Earth's carbon.

#### ***E-ESS2-6 Academic Language***

##### Question/Sentence Stems

- One piece of evidence we have for the cycling of matter in this system is \_\_\_\_\_.
- The flow/transfer of energy causes \_\_\_\_\_ to occur in the system.
- Energy transfer mechanisms include \_\_\_\_\_.
- When energy leaves the system, the energy \_\_\_\_\_.

##### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                                     |                  |                         |
|-------------------------------------|------------------|-------------------------|
| • abiotic                           | • cellular       | • inorganic             |
| • accumulate                        | • respiration    | • input                 |
| • anthropogenic impact              | • chemosynthesis | • marine snow           |
| • atmosphere                        | • climate change | • organic               |
| • bacteria                          | • compounds      | • organism              |
| • basin                             | • concentration  | • output                |
| • biogeochemical cycle              | • decay          | • oxygen                |
| • biomass                           | • decompose      | • pathway               |
| • biosphere                         | • decomposition  | • photosynthesis        |
| • biotic                            | • diffusion      | • phytoplankton         |
| • byproduct                         | • element        | • plankton              |
| • carbon                            | • emissions      | • pool                  |
| • carbon cycle                      | • equilibrium    | • rate of transfer/flow |
| • carbon dioxide (CO <sub>2</sub> ) | • flux           | • reservoir             |
| • carbon sink                       | • fungi          | • sediment              |
|                                     | • geosphere      | • sequestered           |
|                                     | • human activity | • soil                  |
|                                     | • hydrocarbon    |                         |
|                                     | • hydrosphere    |                         |

**E-ESS2-7. Communicate scientific information that** illustrates how Earth’s systems and life on Earth *change and influence each other over time*.

**Clarification Statement:** Emphasis is on the dynamic causes, effects, and feedbacks between the biosphere and Earth’s other systems, whereby geoscience factors control the evolution of life, which in turn continuously alters Earth’s surface. Examples could include how photosynthetic life altered the atmosphere through the production of oxygen, which in turn increased weathering rates and allowed for the evolution of animal life; how microbial life on land increased the formation of soil, which in turn allowed for the evolution of land plants; or how the evolution of corals created reefs altered patterns of erosion and deposition along coastlines and provided habitats for the evolution of new life forms.

**State Assessment Boundary:** Assessment does not include a comprehensive understanding of the mechanisms of how the biosphere interacts with all of Earth’s other systems.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Crosscutting Concepts                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Obtaining, Evaluating, and Communicating Information</b></p> <p>Obtaining, evaluating, and communicating information in 9-12 builds on K-8 experiences and progresses to evaluating the validity and reliability of the claims, methods, and designs.</p> <p>Communicate scientific and/or technical information or ideas (e.g. about phenomena) in multiple formats (i.e., orally, graphically, textually, mathematically).</p> <p>Use words, tables, diagrams, and graphs, as well as mathematical expressions to communicate their understanding or to ask questions about a system under study.</p> | <p><b>ESS2.D: Weather and Climate</b></p> <p>Gradual atmospheric changes were due to plants and other organisms that captured carbon dioxide and released oxygen. The time scales of these changes varied from a few to millions of years.</p> <p><b>ESS2.E: Biogeology</b></p> <p>As Earth changes, life on Earth adapts and evolves to those changes, so just as life influences other Earth systems, other Earth systems influence life. Life and the planet’s nonliving systems can be said to co-evolve. The many dynamic and delicate feedbacks between the biosphere and other Earth systems cause a continual co-evolution of Earth’s surface and the life that exists on it.</p> | <p><b>Stability and Change</b></p> <p>Much of science deals with constructing explanations of how things change and how they remain stable.</p> |

Observable features of student performance by the end of the course:

**1. Communication**

- a. Students use and cite at least two different formats (for example: oral, graphical, textual, mathematical, etc.) to communicate scientific information about the coevolution of Earth's systems and life on Earth.

**2. Connections**

- a. Students identify and communicate evidence for the coevolution of Earth's systems and life on Earth, including:
  - i. scientific explanations about the composition of Earth's atmosphere shortly after its formation,
  - ii. current atmospheric conditions,
  - iii. evidence for the emergence of photosynthetic organisms,
  - iv. evidence for the effect of the presence of free oxygen on evolution and processes in other Earth systems, and
  - v. other evidence that changes in the biosphere affect other Earth systems.
- b. Students identify and communicate that the evolution of photosynthetic organisms led to a drastic change in Earth's atmosphere and oceans in which the free oxygen produced caused worldwide deposition of iron oxide formations, increased weather due to an oxidizing atmosphere, and the evolution of life that depends on oxygen for respiration.

***E-ESS2-7 Academic Language***

Question/Sentence Stems

- The things that stay the same are \_\_\_\_\_.
- The things that change are \_\_\_\_\_.
- The things that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_ (event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing this system to be unstable.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- atmosphere
- atmospheric change
- atmospheric composition
- atmospheric conditions
- biosphere
- biotic
- carbon cycle
- carbon dioxide
- change
- coastline
- coevolution
- continental boundary
- continental shelf
- convection
- coral reef
- crustal deformation
- deposition
- dynamic
- erosion
- evolution
- feedback
- fossils
- fracture zone
- geologic evidence
- geologic time scale
- geoscience
- geosphere
- greenhouse gas
- groundwater
- habitat
- hydrosphere
- igneous rock
- mass extinction
- mass wasting
- metamorphic rock
- microbial
- molten rock
- ocean basin
- ocean trench
- organism
- oxidizing atmosphere
- oxygen
- oxygen-rich
- photosynthesis
- photosynthetic organisms
- plate movement
- plate tectonics
- radioactive
- respiration
- rock formation
- rock strata
- sedimentary rock
- sedimentation
- simultaneous
- soil
- stability
- surface runoff
- time scale
- water cycle
- weather

## ESS3 – Earth and Human Activity

**E-ESS3-1. Construct an explanation based on evidence for how** the availability of natural resources and occurrence of natural hazards *have influenced human activity*.

**Clarification Statement:** Examples of key natural resources could include access to fresh water (such as rivers, lakes, and groundwater), regions of fertile soils such as river deltas, and high concentrations of minerals and fossil fuels. Examples of natural hazards can be from interior processes (such as volcanic eruptions and earthquakes), surface processes (such as tsunamis, mass wasting, and soil erosion), and severe weather (such as hurricanes, floods, and droughts).

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Crosscutting Concepts                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Constructing Explanations and Designing Solutions</b></p> <p>Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific knowledge, principles, and theories.</p> <p>Apply scientific ideas, principles, and/or evidence to provide an explanation of phenomena and solve design problems, taking into account possible unanticipated effects.</p> | <p><b>ESS3.A: Natural Resources</b></p> <p>Resource availability has guided the development of human society.</p> <p><b>ESS3.B: Natural Hazards</b></p> <p>Natural hazards and other geologic events have shaped the course of human history by destroying buildings and cities, eroding land, changing the course of rivers, and reducing the amount of arable land. These have significantly altered the sizes of human populations and have driven human migrations. Natural hazards can be local, regional, or global in origin, and their risks increase as populations grow.</p> <p><b>ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World</b></p> <p>Modern civilization depends on major technological systems. (<i>secondary</i>)</p> | <p><b>Cause and Effect</b></p> <p>Empirical evidence is required to differentiate between cause and correlations and make claims about specific causes and effects.</p> |

Observable features of student performance by the end of the course:

**1. Articulating the explanations of phenomena**

- a. Students articulate a statement describing/explaining the relationship between human activity and environmental factors including:
  - i. relationships between environmental factors (natural hazards, natural resource availability) and features of human societies (for example: population size, migration patterns, etc.), and
  - ii. that technology in modern civilization has mitigated some of the effects of environmental factors on human activity.

**2. Evidence**

- a. Students identify and describe the evidence to construct the explanation, including:
  - i. natural hazard occurrences that can affect human activity and have significantly altered the sizes and distributions of human populations in particular regions,
  - ii. features of human societies that have been affected by the availability of natural resources, and
  - iii. evidence of the dependence of human populations on technological systems to acquire natural resources and to modify physical settings.

**3. Reasoning**

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
  - i. effects of natural hazards and natural resource availability on features of human societies, and
  - ii. how technology has changed the cause-and-effect relationship between the development of human society and environmental factors.

## ***E-ESS3-1 Academic Language***

### Question/Sentence Stems

- By looking at patterns in the data, I/we determined that \_\_\_\_\_ caused \_\_\_\_\_.
- \_\_\_\_\_ caused the patterns I am observing. I know this because \_\_\_\_\_.
- If \_\_\_\_\_ happens, I/we predict that \_\_\_\_\_ will occur.
- The evidence \_\_\_\_\_ presented in the scenario supports the claim that \_\_\_\_\_ causes \_\_\_\_\_.
- In order to conclude that \_\_\_\_\_ caused \_\_\_\_\_, the following evidence is needed \_\_\_\_\_.
- \_\_\_\_\_ is an example of a natural resource that humans depend on, because \_\_\_\_\_.
- The distribution of human communities is dependent on the \_\_\_\_\_ of natural resources.
- Scarcity of natural resources will result in \_\_\_\_\_.
- Climate change will negatively affect human activity because \_\_\_\_\_.
- Natural hazards will negatively affect human activity because \_\_\_\_\_.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                  |                    |                     |
|------------------|--------------------|---------------------|
| • arable land    | • human activity   | • population growth |
| • atmosphere     | • hurricane        | • population size   |
| • availability   | • hydrosphere      | • renewable         |
| • biosphere      | • interior process | • resource          |
| • climate change | • landslide        | • availability      |
| • destruction    | • magnitude        | • risk              |
| • drought        | • mass wasting     | • river course      |
| • earthquake     | • migration        | • river delta       |
| • economic cost  | • mineral          | • scarcity          |
| • erosion        | • concentration    | • severe weather    |
| • fertile soil   | • mitigation       | • soil erosion      |
| • flood          | • modern           | • surface process   |
| • fossil fuels   | • civilization     | • technology        |
| • frequency      | • natural hazard   | • tsunami           |
| • freshwater     | • natural resource | • volcanic eruption |
| • groundwater    | • non-renewable    | • wildfire          |
| • heatwave       | • optimal          |                     |

**E-ESS3-2. Evaluate competing design solutions that address *the impacts of* developing, managing, and using Earth’s energy and mineral resources.**

**Clarification Statement:** Emphasis is on the conservation, recycling, and reuse of resources (such as minerals and metals) where possible, and on minimizing impacts where it is not. Examples include developing best practices for agricultural soil use, mining, forestry, and risk/benefit analysis of the production of conventional, unconventional, or renewable energy resources.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Crosscutting Concepts                                                                                                                                                                                                           |
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| <p><b>Constructing Explanations and Designing Solutions</b></p> <p>Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific knowledge, principles, and theories.</p> <p>Design, evaluate, and/or refine a solution to a complex real-world problem, based on scientific knowledge, student generated sources of evidence, prioritized criteria, and trade off considerations.</p> | <p><b>ESS3.A: Natural Resources</b></p> <p>All forms of energy production and other resource extraction have associated economic, social, environmental, and geopolitical costs and risks, as well as benefits. New technologies and social regulations can change the balance of these factors.</p> <p><b>ETS1.A: Defining and Delimiting an Engineering Problem</b></p> <p>Humanity faces major global challenges today, such as the need for supplies of clean water and food or for energy sources that minimize pollution, which can be addressed through engineering. These global challenges also may have manifestations in local communities.</p> <p><b>ETS1.B: Developing Possible Solutions</b></p> <p>When evaluating solutions, it is important to take into account a range of constraints, including cost, safety, reliability, and aesthetics, and to consider social, cultural, and environmental impacts. Testing should lead to improvements in the design through an iterative procedure. <i>(secondary)</i></p> <p><b>ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World</b></p> <p>Engineers continuously modify these technological systems by applying scientific knowledge and engineering design practices to increase benefits while decreasing costs and risks. Analysis of costs and benefits is a critical aspect of decisions about technology. <i>(secondary)</i></p> | <p><b>Cause and Effect</b></p> <p>Cause-and-effect relationships can be suggested and predicted for complex natural and human designed systems by examining what is known about smaller scale mechanisms within the system.</p> |

Observable features of student performance by the end of the course:

**1. Identifying the design solution**

- a. Students identify and describe the problem each design solution addresses.
- b. Students identify the solution that has the most preferred cost-benefit ratios.

**2. Identifying evidence**

- a. Students identify and describe the evidence for the design solutions, including:
  - i. societal needs for that energy and/or mineral resource,
  - ii. extraction and/or development cost,
  - iii. costs and benefits of the design solution, and
  - iv. feasibility, costs, and benefits of recycling and/or reusing the mineral resource, if applicable.

**3. Evaluating and critiquing the design solution**

- a. Students evaluate the design solutions, including:
  - i. relative strengths and weaknesses of the design solution, based on economic, environmental, and geopolitical costs, risks, and benefits,
  - ii. reliability and validity of the evidence used to evaluate, and
  - iii. constraints, including:
    - 1. cost,
    - 2. safety,
    - 3. reliability,
    - 4. aesthetics,
    - 5. cultural effects, and
    - 6. environmental effects.

**4. Reasoning and synthesis**

- a. Students use logical arguments based on the evaluation of the design solutions, costs and benefits, empirical evidence, and scientific understanding to support one design over the other(s).
- b. Students acknowledge that a decision on the “best” solution may change over time as engineers and scientists work to increase the benefits of design solutions while decreasing cost and risk.

## ***E-ESS3-2 Academic Language***

### Question/Sentence Stems

- By looking at patterns in the data, I/we determined that \_\_\_\_\_ caused \_\_\_\_\_.
- \_\_\_\_\_ caused the patterns I am observing. I know this because \_\_\_\_\_.
- If \_\_\_\_\_ happens, I/we predict that \_\_\_\_\_ will occur.
- The evidence \_\_\_\_\_ presented in the scenario supports the claim that \_\_\_\_\_ causes \_\_\_\_\_.
- In order to conclude that \_\_\_\_\_ caused \_\_\_\_\_, the following evidence is needed \_\_\_\_\_.
- \_\_\_\_\_ is an example of a natural resource that humans depend on, because \_\_\_\_\_.
- The distribution of human communities is dependent on the \_\_\_\_\_ of natural resources.
- Scarcity of natural resources will result in \_\_\_\_\_.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                         |                    |                           |
|-------------------------|--------------------|---------------------------|
| • abiotic               | • extraction       | • recycle                 |
| • aesthetics            | • feasibility      | • regulation              |
| • agriculture           | • feedback         | • reliability             |
| • aquifer               | • fertile land     | • renewable               |
| • basin                 | • forestry         | • resource management     |
| • biotic                | • fossil fuel      | • reuse                   |
| • conservation          | • fragmentation    | • reversible              |
| • constraints           | • groundwater      | • risk/benefit analysis   |
| • consumption           | • hydrothermal     | • safety                  |
| • conventional resource | • irreversible     | • social impact           |
| • cost-benefit analysis | • logarithmic      | • societal                |
| • criteria              | • manufacturing    | • solar radiation         |
| • depletion             | • metal            | • species                 |
| • development           | • mineral resource | • stabilize               |
| • domestic use          | • mining           | • sustainability          |
| • dynamic               | • mitigation       | • tar sand                |
| • economic cost         | • modification     | • technology              |
| • economic impact       | • non-renewable    | • timber                  |
| • economics             | • oil shale        | • unconventional resource |
| • empirical evidence    | • optimal solution | • urban planning          |
| • environmental impact  | • per-capita       | • waste management        |
| • exponential           | • pollution        | • wetland                 |
| • extinction            | • production       |                           |

**E-ESS3-3. Use computational representation to illustrate** the relationships among the management of Earth’s resources, the *sustainability of human populations*, and biodiversity.

**Clarification Statement:** Examples of factors that affect the management of natural resources could include costs of resource extraction and waste management, per-capita consumption, and the development of new technologies. Examples of factors that affect human sustainability could include agricultural efficiency, levels of conservation, urban planning, as well as local and international policies.

**State Assessment Boundary:** Assessment for computational thinking is limited to using provided multi-parameter programs or constructing simplified spreadsheet calculations.

| <b>Science and Engineering Practices</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Crosscutting Concepts</i>                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <p><b>Using Mathematics and Computational Thinking</b></p> <p>Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Simple computational simulations are created and used based on mathematical models of basic assumptions.</p> <p>Use a computational representation of phenomena or design solutions to describe and/or support claims and/or explanations.</p> | <p><b>ESS3.C: Human Impacts on Earth Systems</b></p> <p>The sustainability of human societies and the biodiversity that supports them requires responsible management of natural resources. When the source of an environmental problem is understood and international agreement can be reached, human activities can be regulated to mitigate global impacts (e.g., acid rain and the ozone hole near Antarctica).</p> <p><b>ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World</b></p> <p>Modern civilization depends on major technological systems. <i>(secondary)</i></p> <p>New technologies can have deep impacts on society and the environment, including some that were not anticipated. <i>(secondary)</i></p> | <p><b>Stability and Change</b></p> <p>Feedback (negative or positive) can stabilize or destabilize a system.</p> |

Observable features of student performance by the end of the course:

**1. Representation**

- a. Students develop/use a computational model (for example: spreadsheet, multi-parameter program) that includes representations of the relevant components of an identified ecosystem, including:
  - i. natural resource,
  - ii. sustainability of the human population,
  - iii. biodiversity, and
  - iv. the effect of technology.

**2. Mathematical and/or computational modeling**

- a. Students develop/use a computational model to describe simplified realistic (corresponding to real-world data) relationships between variables to indicate and understanding of the factors (for example: cost, technology, etc.) that affect the management of natural resources, human sustainability, and biodiversity, for example:
  - i. A relationship could be described that the amount of a natural resource does not affect the sustainability of human populations in a given ecosystem without appropriate technology that makes use of the resource; or a relationship could be described that if a given ecosystem is not able to sustain biodiversity, its ability to sustain a human population is also small.
- b. Students develop/use a computational model (for example: spreadsheet, multi-parameter program) that models each component and its simplified mathematical relationship to other components, for example:
  - i.  $S = C \cdot B \cdot R \cdot T$  where:
    - 1. S is sustainability of human populations,
    - 2. C is a constant,
    - 3. B is biodiversity,
    - 4. R is the natural resource, and
    - 5. T is a technology used to extract the resource, so that:
      - a. if there is zero natural resource ( $R=0$ ), zero technology to extract that resource ( $T=0$ ), or zero biodiversity ( $B=0$ ), the sustainability of human populations is also zero, and
  - ii.  $B = B_1 + C \cdot T$  where:
    - 1. B is biodiversity,
    - 2.  $B_1$  is a constant baseline biodiversity,
    - 3. C is a constant that expresses the effect of technology, and
    - 4. T is a technology, so that:
      - a. a technology could either increase or decrease biodiversity depending on the value of C.

- c. Students develop/use a computational model that contains user-controlled variable that can illustrate relationships among the components.

### **3. Analysis**

- a. Students develop/use a computational model and use the results to:
  - i. Illustrate the effect of one component by altering other components in the system or the relationship between components.
  - ii. Identify the effects of technology on the interactions between human populations, natural resources, and biodiversity.
  - iii. Identify feedbacks between the components and whether the feedback stabilizes or destabilizes the system.
- b. Students compare the results to real-world example(s) and determine if the model can be viewed as realistic.
- c. Students identify and describe limitations of the model.

### ***E-ESS3-3 Academic Language***

#### **Question/Sentence Stems**

- The data shows that \_\_\_\_\_ changes over time.
- \_\_\_\_\_ supports the idea that \_\_\_\_\_ has changed over geologic time.
- The components that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_(event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing/caused this system to be unstable.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- acid rain
- agriculture
- anthropogenic impact
- aquifer
- baseline
- basin
- biodiversity ( $B=B1+CT$ )
- biome
- biotic
- conservation
- constant
- consumption
- cost
- depletion
- destabilize
- dynamic
- ecological
- economic cost
- ecosystem
- efficiency
- energy production
- environmental cost
- environmental problem
- exponential
- extinction
- extract
- feedback
- fertile land
- fragmentation
- geopolitical cost
- geosphere
- global impact
- human activity
- human population
- hydrologic cycle
- hydrosphere
- hydrothermal
- irreversible
- logarithmic
- manufacturing
- mineral
- mining
- mitigation
- modern civilization
- natural resource
- non-renewable
- oil shale
- ozone hole
- per-capita
- recycle
- regulation
- renewable
- resource extraction
- resource management
- reuse
- reversible
- societal
- societal cost
- solar radiation
- stabilize
- sustainability ( $S=CBRT$ )
- tar sand
- technology
- timber
- urban planning
- waste management
- watershed
- wetland

**E-ESS3-4. Evaluate or refine a technological solution** *that reduces impacts of human activities on natural systems.*

**Clarification Statement:** Examples of data on the impacts of human activities could include the quantities and types of pollutants released, changes to biomass and species diversity, or areal changes in land surface use (such as for urban development, agriculture and livestock, or surface mining). Examples for limiting future impacts could range from local efforts (such as reducing, reusing, and recycling resources) to large-scale geoengineering design solutions (such as altering global temperatures by making large changes to the atmosphere or ocean).

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Crosscutting Concepts                                                                                            |
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| <p><b>Constructing Explanations and Designing Solutions</b></p> <p>Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific knowledge, principles, and theories.</p> <p>Design or refine a solution to a complex real- world problem, based on scientific knowledge, student-generated sources of evidence, prioritized criteria, and trade off considerations.</p> | <p><b>ESS3.C: Human Impacts on Earth Systems</b></p> <p>Scientists and engineers can make major contributions by developing technologies that produce less pollution and waste and that preclude ecosystem degradation.</p> <p><b>ETS1.B: Developing Possible Solutions</b></p> <p>When evaluating solutions, it is important to take into account a range of constraints, including cost, safety, reliability, and aesthetics, and to consider social, cultural, and environmental impacts. Testing should lead to improvements in the design through an iterative procedure.</p> <p><b>EST2.B: Influence of Engineering, Technology, and Science on Society and the Natural World</b></p> <p>Engineers continuously modify these technological systems by applying scientific knowledge and engineering design practices to increase benefits while decreasing costs and risks.</p> | <p><b>Stability and Change</b></p> <p>Feedback (negative or positive) can stabilize or destabilize a system.</p> |

Observable features of student performance by the end of the course:

**1. Identifying the design solution and supporting evidence**

- a. Students use scientific information to generate a number of possible refinements to a technological solution. Students:
  - i. describe the system being impacted and how human activity is affecting that system,
  - ii. identify the scientific knowledge and reasoning on which the solution is based,
  - iii. describe how the solution functions and may be stabilizing or destabilizing the natural system, and
  - iv. refine the solution that reduces human impact on natural systems.

**2. Identifying additional relevant evidence**

- a. Students evaluate the solution for:
  - i. cost,
  - ii. aesthetics,
  - iii. its impact on overall environmental stability and changes,
  - iv. reliability,
  - v. safety, and/or
  - vi. social, cultural, and environmental impacts.

**3. Evaluating and critiquing the design solution**

- a. Students describe how the refinement will improve the solution to increase benefits and/or decrease costs or risks to people and the environment.
- b. Students evaluate the proposed refinements for:
  - i. effects on the overall stability of and changes in natural systems,
  - ii. cost,
  - iii. safety,
  - iv. aesthetics,
  - v. reliability, and
  - vi. social, cultural, and environmental impacts.

***E-ESS3-4 Academic Language***

Question/Sentence Stems

- The things that stay the same are \_\_\_\_\_.
- The things that change are \_\_\_\_\_.
- The things that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_(event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing this system to be unstable.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- aesthetics
- agriculture
- aquifer
- areal changes
- atmosphere
- availability
- basin
- biodiversity
- biomass
- biosphere
- biotic
- conservation
- constraints
- consumption
- criteria
- depletion
- destabilize
- development
- domestic use
- dynamic
- economic cost
- economic impact
- economics
- ecosystem degradation
- environmental impact
- exponential
- extinction
- extract
- feedback
- fertile land
- fragmentation
- geoengineering
- geosphere
- global temperature
- human activity
- hydrologic cycle
- hydrosphere
- hydrothermal
- impact reduction
- irreversible
- land surface use
- large-scale
- livestock
- logarithmic
- manufacturing
- metal
- mining
- mitigation
- non-renewable
- oil shale
- optimal solution
- per-capita
- pollutant
- production
- recycle
- reduce
- refine
- regulation
- reliability
- renewable
- reuse
- reversible
- scarcity
- societal
- societal impact
- solar radiation
- species diversity
- stabilize
- tar sand
- technology
- timber
- urban development
- urban planning
- waste management
- watershed
- wetland

**E-ESS3-5. Analyze data** and the results from global climate models **to make an evidence-based forecast** of the current rate of regional or global climate change and associated future impacts to Earth’s systems.

**Clarification Statement:** Examples of evidence, for both data and climate model outputs, are for climate changes (such as precipitation and temperature) and their associated impacts (such as on sea level, glacial ice volumes, or atmosphere and ocean composition).

**State Assessment Boundary:** Assessment is limited to one example of a climate change and its associated impacts.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                              | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                   | Crosscutting Concepts                                                                                                                                                                  |
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| <p><b>Analyzing and Interpreting Data</b></p> <p>Analyzing data in 9-12 builds on K-8 experiences and progresses to introducing more detailed statistical analysis, the comparison of data sets for consistency, and the use of models to generate and analyze data.</p> <p>Analyze data using computational models in order to make valid and reliable scientific claims.</p> | <p><b>ESS3.D: Global Climate Change</b></p> <p>Global climate models are often used to understand the process of climate change because these changes are complex and can occur slowly over Earth’s history.</p> <p>Though the magnitudes of human impacts are greater than they have ever been, so too are human abilities to model, predict, and manage current and future impacts.</p> | <p><b>Stability and Change</b></p> <p>Change and rates of change can be quantified and modeled over very short or very long periods of time. Some system changes are irreversible.</p> |

Observable features of student performance by the end of the course:

### 1. Organizing data

- a. Students organize data from global climate models and climate observations over time that relate to the effect of climate change on the physical parameters and/or chemical composition of Earth systems, including the:
  - i. atmosphere,
  - ii. geosphere,
  - iii. hydrosphere, and/or
  - iv. cryosphere.

## **2. Identifying relationships**

- a. Students analyze data to determine the relationships within the datasets, including:
  - i. changes over time on multiple scales and
  - ii. relationships between quantities.

## **3. Interpreting data**

- a. Students use the analyzed data to explain an aspect of past or present climate and the associated physical parameter (for example: temperature, precipitation, sea level, etc.) and/or chemical composition (for example: ocean pH, etc.) of a system.
- b. Students use the analyzed data to predict the future effect of an aspect of climate change on physical parameters and/or chemical composition of a system.
  - i. Students describe whether the predicted effect on the system is reversible or irreversible.
  - ii. Students identify and describe one source of uncertainty in the prediction.
- c. Students include the following in the interpretation of the data:
  - i. make a statement regarding how variation or uncertainty in the data (for example: limitations, accuracy, bias, scale, instrumentation, etc.) may affect the interpretation and
  - ii. identify the limitations of the models that provided the simulation data and ranges for their predictions.

### ***E-ESS3-5 Academic Language***

#### **Question/Sentence Stems**

- The things that stay the same are \_\_\_\_\_.
- The things that change are \_\_\_\_\_.
- The things that are changing slowly/quickly in this system are \_\_\_\_\_.
- The \_\_\_\_\_(event) changed this system by \_\_\_\_\_.
- \_\_\_\_\_ was affected by the change of \_\_\_\_\_.
- \_\_\_\_\_ are causing this system to be unstable.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- accuracy
- acid rain
- anthropogenic
- atmosphere
- atmospheric change
- atmospheric composition
- average air temperature
- average sea surface temperature
- biosphere
- carbon cycle
- carbon footprint
- climate change
- concentration
- cryosphere
- electromagnetic radiation
- extinction
- forecast
- geochemical reaction
- geoscience
- geosphere
- glacial ice volume
- glacier
- global climate records
- global temperature
- global warming
- greenhouse effect
- greenhouse gas
- human impact
- hydrosphere
- ice core
- input
- instrumentation
- irreversible
- limitation
- methane
- Milankovitch cycle
- ocean consumption
- orientation
- output
- pH
- precipitation
- probabilistic
- radiation
- redistribute
- reversible
- sea level
- temperature
- trends
- volcanic ash

**E-ESS3-6. Use a computational representation to illustrate** *the relationships among Earth systems and how those relationships are being modified due to human activity.*

**Clarification Statement:** Examples of Earth systems to be considered are the hydrosphere, atmosphere, cryosphere, geosphere, and/or biosphere. One example of this relationship is how human activities can create changes in the atmosphere including an increase in carbon dioxide that has many far-reaching effects, including changes in photosynthetic biomass on land, ocean acidification, and storm intensity.

**State Assessment Boundary:** Assessment does not include running computational representations and is limited to using the published results of scientific computational models.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Crosscutting Concepts                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Using Mathematics and Computational Thinking</b></p> <p>Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Simple computational simulations are created and used based on mathematical models of basic assumptions.</p> <p>Use a computational representation of phenomena or design solutions to describe and/or support claims and/or explanations.</p> | <p><b>ESS2.D: Weather and Climate</b></p> <p>Current models predict that, although future regional climate changes will be complex and varied, average global temperatures will continue to rise. The outcomes predicted by global climate models strongly depend on the amounts of human-generated greenhouse gases added to the atmosphere each year and by the ways in which these gases are absorbed by the ocean and biosphere. Hence the outcomes depend on human behaviors as well as on natural factors that involve complex feedback among Earth’s systems. (<i>secondary</i>)</p> <p><b>ESS3.D: Global Climate Change</b></p> <p>Through computer simulations and other studies, important discoveries are still being made about how the ocean, the atmosphere, and the biosphere interact and are modified in response to human activities.</p> <p><b>ESS3.B: Natural Hazards</b></p> <p>Human activities can contribute to the frequency and intensity of some natural hazards.</p> | <p><b>Systems and System Models</b></p> <p>When investigating or describing a system, the boundaries and initial conditions of the system need to be defined and their inputs and outputs analyzed and described using models.</p> |

Observable features of student performance by the end of the course:

**1. Representation**

- a. Students develop/use a computational model of interacting Earth systems and describe the relevant components, including:
  - i. relevant system components,
  - ii. system boundaries,
  - iii. initial conditions,
  - iv. inputs and outputs, and
  - v. relationships that determine the interaction (for example: relationship between atmospheric CO<sub>2</sub> and production of photosynthetic biomass and ocean acidification).

**2. Mathematical and/or computational modeling**

- a. Students develop/use a computational model of Earth systems and describe relationships among at least two of Earth's systems, including how the relevant components in each individual Earth system can drive changes in another, interacting Earth system.

**3. Analysis**

- a. Students develop/use a computational model as evidence to describe how human activity could affect the relationships between Earth's systems.

***E-ESS3-6 Academic Language***

Question/Sentence Stems

- The key components of the system are \_\_\_\_\_.
- In the system, \_\_\_\_\_ and \_\_\_\_\_ are shown in the model.
- In the system, \_\_\_\_\_ and \_\_\_\_\_ work together to \_\_\_\_\_.
- In the system, \_\_\_\_\_ and \_\_\_\_\_ interact in \_\_\_\_\_ way.
- If you change \_\_\_\_\_ in the system, \_\_\_\_\_ will occur.
- In the system, \_\_\_\_\_ is not shown in the model. This is not shown because \_\_\_\_\_.
- The key assumptions to the model of my system are \_\_\_\_\_ this affects the reliability of the model because \_\_\_\_\_.

## Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- absorption
- agriculture
- anthropogenic
- atmosphere
- atmospheric change
- atmospheric composition
- average air temperature
- average sea surface temperature
- biodiversity
- biomass
- biosphere
- biotic
- carbon cycle
- carbon dioxide
- carbon footprint
- climate change
- combustion
- concentration
- conservation
- consumption
- economic impact
- electromagnetic radiation
- environmental impact
- extinction
- extract
- feedback
- fertile land
- final
- fossil fuel
- geochemical reaction
- geopolitical impact
- geoscience
- geosphere
- glacier
- global climate records
- global temperature
- greenhouse gas
- human activity
- hydrosphere
- ice core
- initial
- input
- manufacturing
- metal
- methane
- Milankovitch cycle
- mining
- modification
- natural hazard
- non-renewable
- ocean acidification
- oil shale
- orientation
- output
- per-capita
- photosynthetic biomass
- probabilistic
- radiation
- redistribute
- renewable
- sea level
- societal impact
- tar sand
- technology
- timber
- trends
- volcanic ash
- wetland

**E-ESS3-7. Create an argument, based on evidence that describes how** changes in climate on Earth *have affected* human activity.

**Clarification Statement:** Emphasis is on changes in climate that influence past, modern, or future human activities. Examples of key changes in climate that can affect populations or drive mass migrations could include changes to sea level, regional patterns of temperature and precipitation, and the types of crops and livestock that can be raised.

| Science and Engineering Practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Disciplinary Core Ideas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Crosscutting Concepts                                                                                                                                                  |
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| <p><b>Engaging in Argument from Evidence</b></p> <p>Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about natural and designed world(s). Arguments may also come from current scientific or historical episodes in science.</p> <p>Evaluate the claims, evidence, and reasoning behind currently accepted explanations or solutions to determine the merits of arguments. Construct a scientific argument based on data and evidence.</p> | <p><b>ESS3.D: Global Climate Change</b></p> <p>Impacts of climate change--for example, increased frequency of severe storms due to ocean warming-- have begun to influence human activities. Through computer simulations and other studies, important discoveries are still being made about how the ocean, the atmosphere, and the biosphere interact and are modified in response to human activities, as well as to changes in human activities. Thus, science and engineering will be essential both to understanding the possible impacts of global climate change and to informing decisions about how to slow its rate and consequences for humanity as well as for the rest of the planet.</p> <p>The impacts of climate change are uneven and may affect some regions, species, of human populations more severely than others. By using science-based predictive models, humans can anticipate long-term change more effectively than ever and plan accordingly</p> <p><b>ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World</b></p> <p>Modern civilization depends on major technological systems. (<i>secondary</i>)</p> | <p><b>Cause and Effect</b></p> <p>Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.</p> |

Observable features of student performance by the end of the course:

**1. Developing a claim**

- a. Students make a claim that changes in climate on Earth have affected human activity, including:
  - i. relationships between climate change and features of human societies (for example: population size, migration patterns, etc.), and
  - ii. that technology in modern civilization has mitigated some of the effects of environmental factors on human activity.

**2. Identifying scientific evidence**

- a. Students identify and describe the evidence that supports the claim, including:
  - i. changes in climate (for example: increased incidence and severity of storms, etc.) that can affect human activity (for example: agriculture, etc.) and human populations, and that can drive mass migrations,
  - ii. evidence of the dependence of human populations on technological systems to acquire natural resources and to modify physical settings.

**3. Evaluating and critiquing evidence**

- a. Students identify strengths and weaknesses of the evidence used to support the claim, including:
  - i. types of sources,
  - ii. sufficiency, including validity and reliability, of the evidence to make and defend the claim, and/or
  - iii. any alternative interpretations of the evidence and why the evidence supports or does not support the student's claim, as opposed to any other claims.

**4. Reasoning and synthesis**

- a. Students use the following chain of reasoning to connect the evidence:
  - i. effects of climate change on features of human societies, and
  - ii. how technology has changed the cause-and-effect relationship between the development of human society and environmental factors.
- b. Students defend a claim against counterclaims by evaluating counterclaims and by describing the connections between the relevant and appropriate evidence to the strongest claim.

## ***E-ESS3-7 Academic Language***

### Question/Sentence Stems

- By looking at patterns in the data, I/we determined that \_\_\_\_\_ caused \_\_\_\_\_.
- \_\_\_\_\_ caused the patterns I am observing. I know this because \_\_\_\_\_.
- If \_\_\_\_\_ happens, I/we predict that \_\_\_\_\_ will occur.
- The evidence \_\_\_\_\_ presented in the scenario supports the claim that \_\_\_\_\_ causes \_\_\_\_\_.
- In order to conclude that \_\_\_\_\_ caused \_\_\_\_\_, the following evidence is needed \_\_\_\_\_.
- Climate change will negatively affect human activity because \_\_\_\_\_.

### Terminology to Support Student Discourse about Phenomena

\*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- |                                   |                          |                       |
|-----------------------------------|--------------------------|-----------------------|
| • abiotic                         | • environmental impact   | • mitigate            |
| • agriculture                     | • extinction             | • modern civilization |
| • anthropogenic                   | • extract                | • natural resource    |
| • atmosphere                      | • fertile land           | • non-renewable       |
| • atmospheric change              | • fossil fuel            | • ocean warming       |
| • atmospheric composition         | • geochemical reaction   | • oil shale           |
| • average air temperature         | • geopolitical impact    | • orientation         |
| • average sea surface temperature | • geoscience             | • per-capita          |
| • biodiversity                    | • geosphere              | • precipitation       |
| • biosphere                       | • glacier                | • probabilistic       |
| • biotic                          | • global climate records | • radiation           |
| • carbon cycle                    | • global temperature     | • redistribute        |
| • carbon footprint                | • greenhouse gas         | • regional pattern    |
| • climate change                  | • human activity         | • renewable           |
| • combustion                      | • hydrosphere            | • sea level           |
| • concentration                   | • ice core               | • societal impact     |
| • conservation                    | • manufacturing          | • tar sand            |
| • consumption                     | • mass migration         | • technology          |
| • economic impact                 | • metal                  | • timber              |
| • electromagnetic radiation       | • methane                | • trends              |
|                                   | • Milankovitch cycle     | • volcanic ash        |
|                                   | • mining                 | • wetland             |

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## Appendix A: Earth and Space Science Grade Vertical Articulation of Disciplinary Core Idea Instructional Focus Statements

This appendix presents the vertical articulation of Disciplinary Core Idea (DCI) Aligned Instructional Focus statements found in the Vertical Articulations on the [Science Instructional Resources webpage](#). It is intended to help educators understand how conceptual understanding develops across grade levels and courses and how current learning connects to both prior and future instruction.

This appendix is designed to support instructional planning, curriculum alignment, and understanding of the broader progression of science concepts across grades and courses.

Instructional Focus statements clarify content expectations associated with a Performance Expectation but should be considered alongside the associated Science and Engineering Practices (SEPs) and Crosscutting Concepts (CCCs).

The appendix is organized by Performance Expectation (PE) and associated DCI, using the following format:

### Domain Core Idea

#### *Performance Expectation*

#### Disciplinary Core Idea

| Previous Learning                                                                                                                                                                                                     | Current Course or Content                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Related DCI Aligned Instructional Focus statements from earlier grades are presented in this column to illustrate the conceptual foundations of the current learning.</li></ul> | <ul style="list-style-type: none"><li>DCI Aligned Instructional Focus statements for the current grade are presented in this column. If a statement aligns to more than one PE, the applicable PE code(s) follow the statement. Only relevant Extended Knowledge statements are included.</li></ul> |

The appendix is not intended to represent a complete curriculum or pacing guide. Rather, it provides a conceptual map of how the knowledge and understandings associated with a DCI progress across grades, courses, and content.

## ESS1 – Earth's Place in the Universe

***E-ESS1-1. Develop a model based on evidence to illustrate that energy generated by nuclear fusion within the sun (and other stars) radiates to and influences orbiting planets.***

### ESS1.A: The Universe and Its Stars

| Fifth Grade                                                                                                                                                                                                                                                                                                       | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Identify the Sun as a star observed from Earth.</li><li>Compare the apparent brightness and apparent size of the Sun and other stars as observed from Earth.</li><li>Explain that differences in how stars appear are due to their relative distances from Earth.</li></ul> | <ul style="list-style-type: none"><li>Describe how nuclear fusion within stars releases electromagnetic energy that is observed as starlight. (E-ESS1-2, E-ESS1-3)</li><li>Describe the general stages of stellar lifecycles from formation through burnout. (E-ESS1-2, E-ESS1-3)</li><li>Recognize that starlight and light spectra provide indirect evidence of processes occurring within stars and across the universe. (E-ESS1-2, E-ESS1-3)</li></ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"><li>Recognize that black holes are dense regions formed by the collapse of massive stars, where gravity is strong enough to affect nearby matter and light. (E-ESS1-3)</li></ul> |

### PS3.D: Energy in Chemical Processes and Everyday Life

| Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>State that nuclear fusion in the Sun's core releases energy that is transported outward and reaches Earth as electromagnetic radiation.</li><li>Recognize that changes in solar output can affect the energy input to Earth systems.</li></ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"><li>Recognize that hydrogen fuses to helium in the Sun, producing photons that radiate through space and influence orbiting planets.</li></ul> |

***E-ESS1-2. Construct an explanation of the Big Bang Theory based on evidence to show that the universe is changing over time.***

**ESS1.A: The Universe and Its Stars**

| <b>Eighth Grade</b>                                                                                                                                                                                                                                                                       | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>ESS1.B Earth and the Solar System</p> <ul style="list-style-type: none"> <li>Describe how the solar system is understood to have formed from a disk of gas and dust.</li> <li>Explain that gravity acts as the interaction that holds objects in the solar system in orbit.</li> </ul> | <ul style="list-style-type: none"> <li>Describe how nuclear fusion within stars releases electromagnetic energy that is observed as starlight. (E-ESS1-1, E-ESS1-3)</li> <li>Describe the general stages of stellar lifecycles from formation through burnout. (E-ESS1-1, E-ESS1-3)</li> <li>Identify multiple lines of astronomical evidence that support the Big Bang theory.</li> <li>Relate the formation of elements to processes occurring in stars and during the early history of the universe.</li> <li>Recognize that starlight and light spectra provide indirect evidence of processes occurring within stars and across the universe. (E-ESS1-1, E-ESS1-3)</li> <li>Distinguish between elements formed during the early universe and those formed later through stellar processes. (E-ESS1-3)</li> </ul> |

**PS4.B: Electromagnetic Radiation**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Describe that when light interacts with matter, it may be reflected, absorbed, refracted, or transmitted, depending on the material.</li> <li>Use a wave model of light to explain differences in brightness and color qualitatively.</li> <li>Compare light waves and mechanical waves based on whether a medium is required for travel.</li> </ul> | <ul style="list-style-type: none"> <li>Explain that elements produce characteristic absorption and emission spectra that can be used to identify their presence.</li> <li>Describe how spectral data from electromagnetic radiation are used as evidence to determine the composition of Earth and space systems.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Apply spectroscopic evidence to infer material composition where direct sampling is not possible.</li> </ul> |

## ETS2.A: Interdependence of Science, Engineering, and Technology

| Sixth and Eighth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify a technology that enables new evidence about cells or Earth systems.</li> <li>Classify available tools by the information type each contributes.</li> <li>Explain how a named tool expands accuracy or scope for the scientific concept.</li> </ul> <p>Eighth Grade</p> <ul style="list-style-type: none"> <li>Recognize genetic or remote-sensing advances that opened new questions in life or Earth history.</li> <li>Categorize technologies by the evidence type they add to the same claim.</li> <li>Explain how a named technology raises confidence in a claim about relatedness or geologic time.</li> <li>Analyze which technology better fits a local or global question and state why qualitatively.</li> </ul> | <ul style="list-style-type: none"> <li>Identify instruments and models that make key Earth- and space-science questions answerable. (E-ESS1-4, E-ESS2-3)</li> <li>Classify tools by the evidence they contribute to the same claim. (E-ESS1-4, E-ESS2-3)</li> <li>Explain how one instrument’s data aligns with a key scientific model. (E-ESS1-4, E-ESS2-3)</li> <li>Identify which instrument data (spectrum, image, seismic trace) best supports the model and explain why. (E-ESS1-4, E-ESS2-3)</li> <li>Name one large R&amp;D effort (for example space telescope, seismic array) and the evidence type it adds for a key claim. (E-ESS1-4, E-ESS2-3)</li> <li>Evaluate why a stated R&amp;D collaboration strengthens both the tool and the scientific explanation. (E-ESS1-4, E-ESS2-3)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize one non-science role (for example software, materials) that strengthens the scientific result and describe the basic contribution. (E-ESS1-4, E-ESS2-3)</li> </ul> |

***E-ESS1-3. Construct an explanation using evidence to explain the ways elements are produced over the life cycle of a star.***

ESS1.A: The Universe and Its Stars

| Fifth Grade                                                                                                                                                                                                                                                                                                           | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Identify the Sun as a star observed from Earth.</li> <li>Compare the apparent brightness and apparent size of the Sun and other stars as observed from Earth.</li> <li>Explain that differences in how stars appear are due to their relative distances from Earth.</li> </ul> | <ul style="list-style-type: none"> <li>Describe how nuclear fusion within stars releases electromagnetic energy that is observed as starlight. (E-ESS1-1, E-ESS1-2)</li> <li>Describe the general stages of stellar lifecycles from formation through burnout. (E-ESS1-1, E-ESS1-2)</li> <li>Relate the formation of elements to processes occurring in stars and during the early history of the universe. (E-ESS1-2)</li> <li>Recognize that starlight and light spectra provide indirect evidence of processes occurring within stars and across the universe. (E-ESS1-1, E-ESS1-2)</li> <li>Distinguish between elements formed during the early universe and those formed later through stellar processes. (E-ESS1-2)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that black holes are dense regions formed by the collapse of massive stars, where gravity is strong enough to affect nearby matter and light. (E-ESS1-1)</li> </ul> |

***E-ESS1-4. Use mathematical or computational representations to predict the motion of orbiting objects in the universe due to gravity.***

**ESS1.B: Earth and the Solar System**

| <b>Eighth Grade</b>                                                                                                                                                                                                                                                                                                                                                     | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Identify the major components of the solar system (Earth, Sun, Moon, etc.).</li><li>• Explain how the relative positions of Earth, the Moon, and the Sun result in tides, seasons, moon phases, and eclipses.</li><li>• Explain that gravity acts as the interaction that holds objects in the solar system in orbit.</li></ul> | <ul style="list-style-type: none"><li>• Describe the solar system as a collection of objects of varying sizes and conditions, including the Sun, planets, and smaller bodies.</li><li>• Describe orbital motion in the solar system using Kepler’s laws, including elliptical paths around the Sun.</li><li>• Explain how gravitational interactions and collisions can influence and alter orbits over time.</li></ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"><li>• Recognize that Kepler’s laws describe regular patterns observed across many orbiting systems, not only Earth’s.</li></ul> |

## ETS2.A: Interdependence of Science, Engineering, and Technology

| Sixth and Eighth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify a technology that enables new evidence about cells or Earth systems.</li> <li>Classify available tools by the information type each contributes.</li> <li>Explain how a named tool expands accuracy or scope for the scientific concept.</li> </ul> <p>Eighth Grade</p> <ul style="list-style-type: none"> <li>Recognize genetic or remote-sensing advances that opened new questions in life or Earth history.</li> <li>Categorize technologies by the evidence type they add to the same claim.</li> <li>Explain how a named technology raises confidence in a claim about relatedness or geologic time.</li> <li>Analyze which technology better fits a local or global question and state why qualitatively.</li> </ul> | <ul style="list-style-type: none"> <li>Identify instruments and models that make key Earth- and space-science questions answerable. (E-ESS1-2, E-ESS2-3)</li> <li>Classify tools by the evidence they contribute to the same claim. (E-ESS1-2, E-ESS2-3)</li> <li>Explain how one instrument’s data aligns with a key scientific model. (E-ESS1-2, E-ESS2-3)</li> <li>Identify which instrument data (spectrum, image, seismic trace) best supports the model and explain why. (E-ESS1-2, E-ESS2-3)</li> <li>Name one large R&amp;D effort (for example space telescope, seismic array) and the evidence type it adds for a key claim. (E-ESS1-2, E-ESS2-3)</li> <li>Evaluate why a stated R&amp;D collaboration strengthens both the tool and the scientific explanation. (E-ESS1-2, E-ESS2-3)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize one non-science role (for example software, materials) that strengthens the scientific result and describe the basic contribution. (E-ESS1-2, E-ESS2-3)</li> </ul> |

***E-ESS1-5. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks.***

**ESS1.C: The History of Planet Earth**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Distinguish between relative dating of geologic events and absolute measures of time, without determining numerical ages.</li> <li>Relate differences in rock strata to the sequence in which major geologic events occurred.</li> <li>Interpret patterns in rock layers and fossils as evidence of significant events and changes in Earth's history.</li> <li>Describe how fossil patterns in rock strata show sequences of organism appearance and extinction without explaining biological mechanisms.</li> </ul> | <ul style="list-style-type: none"> <li>Relate the younger age of oceanic crust compared to continental crust as evidence of seafloor creation and destruction.</li> <li>Explain how ongoing geologic processes limit the preservation of Earth's earliest rock record.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that interactions within Earth systems can produce feedback loops, where processes either reinforce change (constructive/positive feedback) or counteract change (destructive/negative feedback) over time. (E-ESS1-6)</li> </ul> |

**ESS2.B: Plate Tectonics and Large-Scale System Interactions**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Explain plate tectonics as the theory that accounts for the movement of Earth's surface rocks over time.</li> <li>Relate plate motion to patterns in the distribution of rocks and fossils within Earth's crust.</li> <li>Explain how plate interactions, such as converging, diverging, or transform, are associated with changes in Earth's surface over geologic time.</li> </ul> | <ul style="list-style-type: none"> <li>Explain plate tectonics as the theory that accounts for the continual reshaping of Earth's surface through constructive and destructive geologic processes.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that Earth's surface continues to change and that the same geologic processes operating today will continue to shape continents into Earth's future. (E-ESS2-1)</li> </ul> |

***E-ESS1-6. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth’s formation and early history.***

**ESS1.B: The Earth and the Solar System**

| <b>Eighth Grade</b>                                                                                                                                                                                                                              | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Describe how the solar system is understood to have formed from a disk of gas and dust.</li> <li>Explain that gravity acts as the interaction that holds objects in the solar system in orbit.</li> </ul> | <ul style="list-style-type: none"> <li>Describe the solar system as a collection of objects of varying sizes and conditions, including the Sun, planets, and smaller bodies. (E-ESS1-4)</li> <li>Explain that the solar system formed when gravity caused a disk of gas and dust to collapse and come together to form the Sun, planets, and other celestial bodies.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Identify that the formation of planetary systems from disks of gas and dust appears to be common in the universe.</li> </ul> |

**ESS1.C: The History of Planet Earth**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <ul style="list-style-type: none"> <li>Describe the geologic time scale as a tool for organizing Earth’s history based on patterns in rock strata and the fossil record.</li> <li>Distinguish between relative dating of geologic events and absolute measures of time, without determining numerical ages.</li> <li>Relate differences in rock strata to the sequence in which major geologic events occurred.</li> <li>Interpret patterns in rock layers and fossils as evidence of significant events and changes in Earth’s history.</li> </ul> | <ul style="list-style-type: none"> <li>Apply principles of relative dating to determine the order of geologic events and the relative ages of rock layers.</li> <li>Relate radioactive decay and isotopic composition in rocks to the establishment of an absolute geologic time scale.</li> <li>Compare Earth’s rock record with rock and material records from other celestial objects to infer Earth’s early history.</li> <li>Describe how surface features observed on other planets and moons provide context for interpreting the sequence of events in Earth’s geologic history.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that interactions within Earth systems can produce feedback loops, where processes either reinforce change (constructive/positive feedback) or counteract change (destructive/negative feedback) over time. (E-ESS1-5)</li> </ul> |

## PS1.C: Nuclear Processes

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### Earth and Space Science

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- Explain that radioactive decay occurs spontaneously and follows predictable, exponential patterns over time.
  - Analyze how known half-lives of radioactive isotopes are used to determine the absolute ages of rocks and other materials.
  - Interpret radiometric age evidence from Earth materials, meteorites, and lunar samples to explain Earth's formation, early history, and the scale of geologic time.
-

## ESS2 – Earth Systems

***E-ESS2-1. Use evidence to argue how Earth’s internal and external processes operate to form and modify continental and ocean-floor features throughout Earth’s history.***

### ESS2.A: Earth Materials and Systems

| Sixth Grade                                                                                                                                                                                                                                                                                                                                                             | Earth and Space Science                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Compare how interactions among Earth’s systems, including those involved in the rock cycle, operate across different spatial and temporal scales.</li><li>• Relate Earth system interactions, including rock formation and transformation, to changes that have shaped Earth’s history and will influence its future.</li></ul> | <ul style="list-style-type: none"><li>• Describe Earth’s systems as dynamic and interacting, with feedback effects that can amplify or reduce change. (E-ESS2-2)</li><li>• Describe how limits in understanding feedbacks among Earth’s systems constrain scientists’ ability to predict some changes and their impacts. (E-ESS2-2)</li></ul> |

### ESS2.B: Plate Tectonics and Large-Scale System Interactions

| Sixth Grade                                                                                                                                                                                                                                                                                                                                                                                                | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Relate plate movement to the formation of major continental and ocean-floor features.</li><li>• Relate plate motion to patterns in the distribution of rocks and fossils within Earth’s crust.</li><li>• Explain how plate interactions, such as converging, diverging, or transform, are associated with changes in Earth’s surface over geologic time.</li></ul> | <ul style="list-style-type: none"><li>• Explain how interactions among tectonic plates contribute to the formation of ocean basins, earthquakes, volcanoes, mountain ranges, and plateaus.</li><li>• Relate mantle convection to the movement of lithospheric plates at Earth’s surface.</li><li>• Use evidence from seafloor spreading, including magnetic patterns in oceanic crust, to support long-term movement of tectonic plates.</li><li>• Describe how radioactive decay within Earth’s interior provides the energy that drives mantle convection.</li></ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"><li>• Recognize that Earth’s surface continues to change and that the same geologic processes operating today will continue to shape continents into Earth’s future. (E-ESS1-5)</li></ul> |

***E-ESS2-2. Analyze data to make the claim that one change to Earth’s surface can create feedbacks that cause changes to other Earth systems.***

**ESS2.A: Earth Materials and Systems**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                |
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| <ul style="list-style-type: none"><li>• Explain how matter cycles through Earth’s systems using the rock cycle as an example of how materials change form over time.</li><li>• Relate energy flow to processes such as melting, weathering, erosion, compaction, and metamorphism that drive changes in Earth’s materials.</li><li>• Compare how interactions among Earth’s systems, including those involved in the rock cycle, operate across different spatial and temporal scales.</li><li>• Relate Earth system interactions, including rock formation and transformation, to changes that have shaped Earth’s history and will influence its future.</li></ul> | <ul style="list-style-type: none"><li>• Describe Earth’s systems as dynamic and interacting, with feedback effects that can amplify or reduce change. (E-ESS2-1)</li><li>• Describe how limits in understanding feedbacks among Earth’s systems constrain scientists’ ability to predict some changes and their impacts. (E-ESS2-1)</li></ul> |

**ESS2.D: Weather and Climate**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Earth and Space Science</b>                                                                                                                                                                                                                            |
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| <ul style="list-style-type: none"><li>• Relate Earth’s axial tilt to seasonal patterns of uneven heating and cooling across the planet.</li><li>• Explain how unequal heating of Earth contributes to large-scale patterns of weather and climate.</li><li>• Relate interactions among sunlight, the ocean, the atmosphere, ice, landforms, and living things to differences in weather and climate patterns.</li><li>• Relate the ocean’s ability to absorb, store, and redistribute energy to its influence on regional weather and climate.</li></ul> | <ul style="list-style-type: none"><li>• Explain Earth’s global climate system as driven by energy from the Sun and its reflection, absorption, storage, redistribution, and re-radiation among the atmosphere, ocean, land, and ice. (E-ESS2-4)</li></ul> |

## ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

| Sixth and Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify how local needs, values, and priorities shape which technologies are used in different places.</li> <li>Classify examples of technology use by regional factors such as climate, available natural resources, and economic conditions.</li> <li>Explain how a new scientific finding can shift which technology is adopted or adapted in a community.</li> <li>Analyze how a limitation (cost, materials, environmental risk) changes whether a technology is practical in the stated context.</li> </ul> <p>Seventh Grade</p> <ul style="list-style-type: none"> <li>Describe how technology use and limits on use vary with community priorities and regional conditions over time.</li> <li>Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment.</li> <li>Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out.</li> <li>Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.</li> </ul> | <ul style="list-style-type: none"> <li>Identify a societal need met by a large-scale Earth/space technology. (E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Classify options by costs, benefits, risks, and trade-offs, noting how these differ by region (climate, resources, economics). (E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Describe how engineers refine technological systems by applying scientific research (climate data, geoscience models, natural-resource analyses) to increase benefits while reducing risks to people and ecosystems. (E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Evaluate which option is most feasible or sustainable for the stated region. (E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Identify one regional factor that most shifts the preferred choice and explain why. (E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> </ul> |

***E-ESS2-3. Develop a model based on evidence of Earth’s interior that describes cycling of matter through convection processes.***

**ESS2.A: Earth Materials and Systems**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>ESS2.A: Earth Materials and Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"><li>• Explain Earth processes by relating energy flow and matter cycling within and among Earth’s systems.</li><li>• Explain how matter cycles through Earth’s systems using the rock cycle as an example of how materials change form over time.</li><li>• Relate energy flow to processes such as melting, weathering, erosion, compaction, and metamorphism that drive changes in Earth’s materials.</li><li>• Compare how interactions among Earth’s systems, including those involved in the rock cycle, operate across different spatial and temporal scales.</li></ul> | <ul style="list-style-type: none"><li>• Describe Earth’s internal structure and the evidence supporting a solid inner core, liquid outer core, solid mantle, and crust.</li><li>• Explain how energy flow from Earth’s interior drives mantle movement and tectonic plate motion by convection currents.</li><li>• Relate transfers of energy and movements of matter to chemical and physical changes in Earth’s materials and living organisms.</li></ul> |
| <b>PS3.B Conservation of Energy and Energy Transfer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"><li>• Demonstrate how thermal energy is transferred spontaneously from hotter objects or regions to colder ones by conduction, convection, and radiation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## ESS2.B: Plate Tectonics and Large-Scale System Interactions

| Sixth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Current Grade/Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <p>ESS2.B: Plate Tectonics and Large-Scale System Interactions</p> <ul style="list-style-type: none"> <li>Explain plate tectonics as the theory that accounts for the movement of Earth’s surface rocks over time.</li> <li>Relate plate movement to the formation of major continental and ocean-floor features.</li> <li>Relate plate motion to patterns in the distribution of rocks and fossils within Earth’s crust.</li> <li>Explain how plate interactions, such as converging, diverging, or transform, are associated with changes in Earth’s surface over geologic time.</li> </ul> <p>PS3.B Conservation of Energy and Energy Transfer</p> <ul style="list-style-type: none"> <li>Demonstrate how thermal energy is transferred spontaneously from hotter objects or regions to colder ones by conduction, convection, and radiation.</li> </ul> | <ul style="list-style-type: none"> <li>Explain plate tectonics as the theory that accounts for the continual reshaping of Earth’s surface through constructive and destructive geologic processes.</li> <li>Explain how interactions among tectonic plates contribute to the formation of ocean basins, earthquakes, volcanoes, mountain ranges, and plateaus. (E-ESS2-1)</li> <li>Relate mantle convection to the movement of lithospheric plates at Earth’s surface. (E-ESS2-1)</li> <li>Describe how radioactive decay within Earth’s interior provides the energy that drives mantle convection. (E-ESS2-1)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that Earth’s surface continues to change and that the same geologic processes operating today will continue to shape continents into Earth’s future. (E-ESS1-5, E-ESS2-1)</li> </ul> |

## PS4.A: Wave Properties

| Sixth and Eighth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"><li>• Identify sound as a mechanical wave that requires a medium to travel.</li><li>• Compare mechanical waves and light waves based on whether a medium is required.</li><li>• Explain that sound waves can be reflected, absorbed, or transmitted when they interact with different materials.</li></ul> <p>Eighth Grade</p> <ul style="list-style-type: none"><li>• Identify a simple wave by describing its wavelength, frequency, and amplitude.</li><li>• Compare wave patterns by examining differences in wavelength, frequency, or amplitude.</li><li>• Explain how changes in amplitude correspond to changes in the energy carried by a wave, qualitatively.</li></ul> | <ul style="list-style-type: none"><li>• Describe how seismic waves travel through Earth and change behavior at boundaries between layers, providing evidence of Earth's interior structure.</li><li>• Compare seismic wave behaviors to infer differences in Earth's internal structure.</li><li>• Explain how reflected and refracted seismic waves provide evidence used to develop models of Earth's interior.</li></ul> |

## ETS2.A: Interdependence of Science, Engineering, and Technology

| Sixth and Eighth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify a technology that enables new evidence about cells or Earth systems.</li> <li>Classify available tools by the information type each contributes.</li> <li>Explain how a named tool expands accuracy or scope for the scientific concept.</li> </ul> <p>Eighth Grade</p> <ul style="list-style-type: none"> <li>Recognize genetic or remote-sensing advances that opened new questions in life or Earth history.</li> <li>Categorize technologies by the evidence type they add to the same claim.</li> <li>Explain how a named technology raises confidence in a claim about relatedness or geologic time.</li> <li>Analyze which technology better fits a local or global question and state why qualitatively.</li> </ul> | <ul style="list-style-type: none"> <li>Identify instruments and models that make key Earth- and space-science questions answerable. (E-ESS1-2, E-ESS1-4)</li> <li>Classify tools by the evidence they contribute to the same claim. (E-ESS1-2, E-ESS1-4)</li> <li>Explain how one instrument’s data aligns with a key scientific model. (E-ESS1-2, E-ESS1-4)</li> <li>Identify which instrument data (spectrum, image, seismic trace) best supports the model and explain why. (E-ESS1-2, E-ESS1-4)</li> <li>Name one large R&amp;D effort (for example space telescope, seismic array) and the evidence type it adds for a key claim. (E-ESS1-2, E-ESS1-4)</li> <li>Evaluate why a stated R&amp;D collaboration strengthens both the tool and the scientific explanation. (E-ESS1-2, E-ESS1-4)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize one non-science role (for example software, materials) that strengthens the scientific result and describe the basic contribution. (E-ESS1-2, E-ESS1-4)</li> </ul> |

***E-ESS2-4. Use a model to describe how causes of short and long-term variations in the flow of energy into and out of Earth’s systems result in changes to climate.***

ESS1.B: Earth and the Solar System

| <b>Eighth Grade</b>                                                                                                                                                                                                                                                                                                                                          | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                          |
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| <ul style="list-style-type: none"><li>• Identify the major components of the solar system (Earth, Sun, Moon, etc.).</li><li>• Explain how the relative positions of Earth, the Moon, and the Sun result in tides, seasons, moon phases, and eclipses.</li><li>• Relate Earth’s axial tilt and orbit to seasonal patterns in different hemispheres.</li></ul> | <ul style="list-style-type: none"><li>• Describe how long-term changes in Earth’s orbital shape and axial orientation alter the intensity and distribution of sunlight reaching Earth.</li><li>• Relate these long-term orbital and rotational variations to cycles of ice ages and gradual long-term changes in climate patterns over geologic time (not human time scales).</li></ul> |

## ESS2.D: Weather and Climate

| Sixth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p>ESS2.D: Weather and Climate</p> <ul style="list-style-type: none"> <li>Explain how unequal heating of Earth contributes to large-scale patterns of weather and climate.</li> <li>Relate interactions among sunlight, the ocean, the atmosphere, ice, landforms, and living things to differences in weather and climate patterns.</li> <li>Describe how latitude, altitude, and geographic features influence atmospheric and oceanic circulation patterns.</li> <li>Relate the ocean's ability to absorb, store, and redistribute energy to its influence on regional weather and climate.</li> </ul> <p>ESS2.C The Role of Water in Earth's Surface Processes</p> <ul style="list-style-type: none"> <li>Relate wind, landforms, and ocean temperatures to patterns of water movement in the atmosphere.</li> <li>Relate patterns of atmospheric water movement to differences in local weather conditions.</li> <li>Explain how variations in temperature and salinity affect water density and contribute to global patterns of interconnected ocean currents.</li> </ul> | <ul style="list-style-type: none"> <li>Explain Earth's global climate system as driven by energy from the Sun and its reflection, absorption, storage, redistribution, and re-radiation among the atmosphere, ocean, land, and ice. (E-ESS2-2)</li> <li>Explain long-term changes in climate patterns as significant and persistent shifts in average or extreme weather conditions over long periods of time.</li> <li>Distinguish between rapid changes in climate patterns caused by sudden natural events and gradual changes in climate patterns occurring over longer timescales. (E-ESS2-6, E-ESS2-7)</li> <li>Explain how geological evidence provides information about past changes in climate patterns across Earth's history. (E-ESS2-6, E-ESS2-7)</li> <li>Relate long-term atmospheric patterns to biological processes and factors that influence the composition of the atmosphere. (E-ESS2-6, E-ESS2-7, E-SS3-6)</li> <li>Explain how increases in greenhouse gases affect Earth's capacity to retain energy. (E-ESS2-6, E-ESS2-7)</li> <li>Describe how future climate patterns are influenced by interactions among Earth systems as well as human activities. (E-ESS2-6, E-ESS2-7, E-ESS3-6)</li> <li>Identify that feedbacks among Earth systems can amplify or reduce changes in climate patterns. (E-ESS2-7, E-ESS3-6)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that long-term changes in climate patterns occur over a wide range of timescales, from years to millions of years. (E-ESS2-6, E-ESS2-7, E-ESS3-6)</li> </ul> |

***E-ESS2-5. Investigate the ways that water (given its unique physical and chemical properties) impacts various Earth systems.***

**ESS2.C: The Roles of Water in Earth’s Surface Processes**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>ESS2.C: The Roles of Water in Earth’s Surface Processes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <ul style="list-style-type: none"> <li>• Explain how the movement of water on land and underground contributes to weathering and erosion that change Earth’s surface features and create underground formations.</li> <li>• Explain how water cycles among land, oceans, and the atmosphere, including changes in form and location over time.</li> <li>• Identify sunlight and gravity as the primary factors that drive water movement and cycling on Earth.</li> <li>• Explain how variations in temperature and salinity affect water density and contribute to global patterns of interconnected ocean currents.</li> </ul> | <ul style="list-style-type: none"> <li>• Relate water’s ability to absorb, store, and release energy during changes of state to its influence on Earth systems, including the biosphere.</li> <li>• Describe how physical properties of water, including expansion upon freezing and the ability to dissolve and transport materials, contribute to weathering, erosion, and other surface changes.</li> <li>• Explain how water’s thermal properties moderate temperature variations across Earth’s surface and within the oceans.</li> </ul> |
| <b>PS1.A Structure and Properties of Matter</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <ul style="list-style-type: none"> <li>• Describe how differences in particle spacing and motion account for the properties of solids, liquids, and gases.</li> <li>• Explain how adding or removing thermal energy changes particle motion, affecting temperature and resulting in changes of state in a pure substance.</li> </ul>                                                                                                                                                                                                                                                                                             | <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>• Identify that relatively small changes in water distribution or state can have large impacts on Earth system behavior over time.</li> </ul>                                                                                                                                                                                                                                                                                                                          |

***E-ESS2-6. Develop a quantitative model to describe the cycling of carbon through the hydrosphere, atmosphere, geosphere, and biosphere.***

**ESS2.D: Weather and Climate**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>ESS2.D: Weather and Climate</p> <ul style="list-style-type: none"> <li>Explain how unequal heating of Earth contributes to large-scale patterns of weather and climate.</li> <li>Relate interactions among sunlight, the ocean, the atmosphere, ice, landforms, and living things to differences in weather and climate patterns.</li> <li>Describe how latitude, altitude, and geographic features influence atmospheric and oceanic circulation patterns.</li> <li>Relate the ocean's ability to absorb, store, and redistribute energy to its influence on regional weather and climate.</li> </ul> <p>ESS2.C The Role of Water in Earth's Surface Processes</p> <ul style="list-style-type: none"> <li>Relate wind, landforms, and ocean temperatures to patterns of water movement in the atmosphere.</li> <li>Relate patterns of atmospheric water movement to differences in local weather conditions.</li> <li>Explain how variations in temperature and salinity affect water density and contribute to global patterns of interconnected ocean currents.</li> </ul> | <ul style="list-style-type: none"> <li>Distinguish between rapid changes in climate patterns caused by sudden natural events and gradual changes in climate patterns occurring over longer timescales. (E-ESS2-4, E-ESS2-7)</li> <li>Explain how geological evidence provides information about past changes in climate patterns across Earth's history. (E-ESS2-4, E-ESS2-7)</li> <li>Relate long-term atmospheric patterns to biological processes and factors that influence the composition of the atmosphere. (E-ESS2-4, E-ESS2-7, E-ESS3-6)</li> <li>Explain how increases in greenhouse gases affect Earth's capacity to retain energy. (E-ESS2-4, E-ESS2-7)</li> <li>Describe how future climate patterns are influenced by interactions among Earth systems as well as human activities. (E-ESS2-4, E-ESS2-7, E-ESS3-6)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that long-term changes in climate patterns occur over a wide range of timescales, from years to millions of years. (E-ESS2-4, E-ESS2-7, E-ESS3-6)</li> </ul> |

***E-ESS2-7. Communicate scientific information that illustrates how Earth’s systems and life on Earth change and influence each other over time.***

**ESS2.D: Weather and Climate**

| <b>Sixth Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p>ESS2.D: Weather and Climate</p> <ul style="list-style-type: none"> <li>Explain how unequal heating of Earth contributes to large-scale patterns of weather and climate.</li> <li>Relate interactions among sunlight, the ocean, the atmosphere, ice, landforms, and living things to differences in weather and climate patterns.</li> <li>Describe how latitude, altitude, and geographic features influence atmospheric and oceanic circulation patterns.</li> <li>Relate the ocean’s ability to absorb, store, and redistribute energy to its influence on regional weather and climate.</li> </ul> <p>ESS2.C The Role of Water in Earth’s Surface Processes</p> <ul style="list-style-type: none"> <li>Relate wind, landforms, and ocean temperatures to patterns of water movement in the atmosphere.</li> <li>Relate patterns of atmospheric water movement to differences in local weather conditions.</li> <li>Explain how variations in temperature and salinity affect water density and contribute to global patterns of interconnected ocean currents.</li> </ul> | <ul style="list-style-type: none"> <li>Distinguish between rapid changes in climate patterns caused by sudden natural events and gradual changes in climate patterns occurring over longer timescales. (E-ESS2-4, E-ESS2-6)</li> <li>Explain how geological evidence provides information about past changes in climate patterns across Earth’s history. (E-ESS2-4, E-ESS2-6)</li> <li>Relate long-term atmospheric patterns to biological processes and factors that influence the composition of the atmosphere. (E-ESS2-4, E-ESS2-6, E-ESS3-4)</li> <li>Explain how increases in greenhouse gases affect Earth’s capacity to retain energy. (E-ESS2-4, E-ESS2-6)</li> <li>Describe how future climate patterns are influenced by interactions among Earth systems as well as human activities. (E-ESS2-4, E-ESS2-6, E-ESS3-6)</li> <li>Identify that feedbacks among Earth systems can amplify or reduce changes in climate patterns. (E-ESS2-4, E-ESS3-4)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that long-term changes in climate patterns occur over a wide range of timescales, from years to millions of years. (E-ESS2-4, E-ESS2-6, E-ESS3-4)</li> </ul> |

## ESS2.E: Biogeology

| Fourth and Eighth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <p>Fourth Grade</p> <ul style="list-style-type: none"> <li>Describe observable effects of vegetation on soil and surface materials.</li> <li>Use evidence to show how plant roots can stabilize soil and reduce erosion.</li> <li>Compare areas with plant cover to areas without plant cover to identify differences in erosion or soil movement.</li> </ul> <p>Eighth Grade</p> <ul style="list-style-type: none"> <li>Relate major environmental changes to patterns of mass extinction observed in Earth's history.</li> <li>Compare the effects of rapid environmental change to slower, long-term change on patterns of life on Earth.</li> <li>Describe how changes in Earth's physical systems and patterns in life on Earth are connected over geologic time.</li> </ul> | <ul style="list-style-type: none"> <li>Describe how changes in Earth's physical systems have influenced the organisms that inhabit the planet over geologic time.</li> <li>Describe how biological activity has influenced Earth's nonliving systems, including the atmosphere, land surface, and oceans.</li> <li>Explain how interactions between the biosphere and other Earth systems contribute to long-term changes in Earth's surface.</li> <li>Explain how feedbacks between life and Earth systems result in the co-evolution of Earth and its life forms over time.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Recognize that feedbacks between life and Earth systems can either amplify or limit environmental change.</li> <li>Distinguish between short-term biological influences on the environment and long-term biogeologic changes.</li> </ul> |

## ESS3 – Earth and Human Activity

***E-ESS3-1. Construct an explanation based on evidence for how the availability of natural resources and occurrence of natural hazards have influenced human activity.***

### ESS3.A: Natural Resources

| Seventh Grade                                                                                                                                                     | Earth and Space Science                                                                                                                                                                     |
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| <ul style="list-style-type: none"><li>Describe how Earth’s natural resources are distributed unevenly across the planet due to past geologic processes.</li></ul> | <ul style="list-style-type: none"><li>Relate differences in resource availability to variations in human settlement patterns, economic activities, and technological development.</li></ul> |

### ESS3.B: Natural Hazards

| Sixth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <ul style="list-style-type: none"><li>Use patterns from the history of natural hazards in a region to explain where similar events are more likely to occur in the future.</li><li>Describe that some natural hazards, such as volcanic eruptions and severe weather, are often preceded by observable warning signs.</li><li>Describe that other natural hazards, such as earthquakes, can occur suddenly with little or no advance notice.</li><li>Explain how Earth processes associated with natural hazards influence the locations and likelihoods of different hazard events.</li></ul> | <ul style="list-style-type: none"><li>Explain how natural hazards and major geologic events have influenced human history by altering landscapes, settlements, and infrastructure.</li><li>Relate natural hazards to changes in rivers, landforms, and the availability of farmable land that affect human activity.</li><li>Explain how natural hazards have influenced human population size, distribution, and migration over time.</li><li>Distinguish between local, regional, and global natural hazards based on their scale and societal impacts.</li><li>Recognize that repeated natural hazards can compound long-term changes in landscapes and human systems.</li></ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"><li>Recognize that the impacts of a single natural hazard can extend far beyond the area where the event occurs.</li></ul> |

## ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

| Sixth and Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify how local needs, values, and priorities shape which technologies are used in different places.</li> <li>Classify examples of technology use by regional factors such as climate, available natural resources, and economic conditions.</li> <li>Explain how a new scientific finding can shift which technology is adopted or adapted in a community.</li> <li>Analyze how a limitation (cost, materials, environmental risk) changes whether a technology is practical in the stated context.</li> </ul> <p>Seventh Grade</p> <ul style="list-style-type: none"> <li>Describe how technology use and limits on use vary with community priorities and regional conditions over time.</li> <li>Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment.</li> <li>Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out.</li> <li>Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.</li> </ul> | <ul style="list-style-type: none"> <li>Identify a societal need met by a large-scale Earth/space technology. (E-ESS2-2, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Classify options by costs, benefits, risks, and trade-offs, noting how these differ by region (climate, resources, economics). (E-ESS2-2, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Describe how engineers refine technological systems by applying scientific research (climate data, geoscience models, natural-resource analyses) to increase benefits while reducing risks to people and ecosystems. (E-ESS2-2, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Evaluate which option is most feasible or sustainable for the stated region. (E-ESS2-2, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Identify one regional factor that most shifts the preferred choice and explain why. (E-ESS2-2, E-ESS3-2, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> </ul> |

***E-ESS3-2. Evaluate competing design solutions that address the impacts of developing, managing, and using Earth’s energy and mineral resources.***

**ESS3.A: Natural Resources**

| <b>Seventh Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <ul style="list-style-type: none"> <li>Identify minerals, freshwater, and biological resources as examples of natural resources used by humans.</li> <li>Relate the availability of certain resources to limits on renewability or replacement over human lifetimes.</li> <li>Distinguish between resources that renew on human timescales and those that form over much longer geologic timescales.</li> <li>Describe how Earth’s natural resources are distributed unevenly across the planet due to past geologic processes.</li> </ul> | <ul style="list-style-type: none"> <li>Explain that the extraction and use of natural resources for energy and materials involve trade-offs, including economic, social, environmental, and geopolitical costs, risks, and benefits.</li> <li>Compare different forms of energy production and resource use in terms of the general types of trade-offs they involve.</li> <li>Describe how technological advances and social regulations can change the balance of costs, risks, and benefits associated with resource use.</li> <li>Distinguish between short-term benefits of resource use and long-term environmental and societal consequences.</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>Identify that impacts of resource extraction and use can extend beyond the location of production.</li> </ul> |

**ETS1.A: Defining and Delimiting an Engineering Problem**

| <b>Previous Learning</b>                                                                                                                                                                                                                                                                                                    | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <ul style="list-style-type: none"> <li><i>ETS1.A does not have a continuous Earth and Space Science progression prior to this. Earlier learning related to defining engineering problems and identifying criteria and constraints is developed primarily through Physical Science instruction in Grades 3–8.</i></li> </ul> | <ul style="list-style-type: none"> <li>Classify an Earth-resource challenge as community level or global.</li> <li>Specify the criteria to be met and the constraints to honor (cost, time, safety; social, cultural, environmental effects).</li> <li>Explain how relevant Earth-system understanding (resource availability, extraction or land-use impacts, renewable vs. non-renewable limits) informs the solution.</li> <li>Analyze the trade-offs among options, including cost-benefit ratios.</li> </ul> |

## ETS1.B: Developing Possible Solutions

| Fourth and Fifth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>Fourth Grade</p> <ul style="list-style-type: none"> <li>• Create a design solution that can be used to gather evidence about how well it solves a problem.</li> <li>• Describe the likely conditions the device or solution must handle.</li> <li>• Compare proposed solutions by matching each to the stated criteria and constraints and noting any mismatch.</li> <li>• Explain which feature(s) most support better performance using grade-appropriate evidence.</li> <li>• Evaluate and communicate one targeted change that would strengthen the next version while staying within the same limits.</li> </ul> <p>Fifth Grade</p> <ul style="list-style-type: none"> <li>• Create a design solution that can be used to gather evidence about how well it solves a problem.</li> <li>• Identify important information about the stated problem from provided sources and name the specific need the solution must address.</li> <li>• Compare proposed solutions by aligning how each meets the given criteria and constraints across the likely conditions.</li> <li>• Identify which criterion (effectiveness, cost, environmental impact) should take top priority for this problem and explain why.</li> <li>• Explain which solution best fits the high-priority criteria qualitatively.</li> <li>• Evaluate one targeted revision that would improve performance in more than one condition while staying within the constraints.</li> </ul> | <ul style="list-style-type: none"> <li>• Identify criteria and constraints for the Earth-resource or impact-mitigation task, including cost, safety, reliability, aesthetics, social/cultural/environmental effects, and environmental stability. (E-ESS3-4)</li> <li>• Compare options by classifying benefits and limitations across economic, environmental, and social dimensions for the same need. (E-ESS3-4)</li> <li>• Explain which criterion should lead the decision and name one key trade-off that is accepted. (E-ESS3-4)</li> <li>• Evaluate a solution refinement that reduces an unintended impact. (E-ESS3-4)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>• Recognize when combining specific features from two options improves fit to the local context and name the pair to keep. (E-ESS3-4)</li> </ul> |

## ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

| Sixth and Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify how local needs, values, and priorities shape which technologies are used in different places.</li> <li>Classify examples of technology use by regional factors such as climate, available natural resources, and economic conditions.</li> <li>Explain how a new scientific finding can shift which technology is adopted or adapted in a community.</li> <li>Analyze how a limitation (cost, materials, environmental risk) changes whether a technology is practical in the stated context.</li> </ul> <p>Seventh Grade</p> <ul style="list-style-type: none"> <li>Describe how technology use and limits on use vary with community priorities and regional conditions over time.</li> <li>Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment.</li> <li>Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out.</li> <li>Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.</li> </ul> | <ul style="list-style-type: none"> <li>Identify a societal need met by a large-scale Earth/space technology. (E-ESS2-2, E-ESS3-1, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Classify options by costs, benefits, risks, and trade-offs, noting how these differ by region (climate, resources, economics). (E-ESS2-2, E-ESS3-1, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Describe how engineers refine technological systems by applying scientific research (climate data, geoscience models, natural-resource analyses) to increase benefits while reducing risks to people and ecosystems. (E-ESS2-2, E-ESS3-1, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Evaluate which option is most feasible or sustainable for the stated region. (E-ESS2-2, E-ESS3-1, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> <li>Identify one regional factor that most shifts the preferred choice and explain why. (E-ESS2-2, E-ESS3-1, E-ESS3-3, E-ESS3-4, E-ESS3-7)</li> </ul> |

***E-ESS3-3. Use computational representation to illustrate the relationships among the management of Earth’s resources, the sustainability of human populations, and biodiversity.***

**ESS3.C: Human Impacts on Earth Systems**

| <b>Seventh Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <ul style="list-style-type: none"> <li>Describe how human activities have altered Earth’s biosphere, including damage to or loss of natural habitats.</li> <li>Relate increases in human population and resource consumption to changes in biodiversity, ecosystem stability, and resource availability.</li> <li>Explain how human-caused environmental changes can have different effects, both negative and positive, on different living things.</li> <li>Describe examples of how monitoring environmental conditions and changing how resources are used can help reduce human impacts.</li> </ul> | <ul style="list-style-type: none"> <li>Describe how the sustainability of human societies depends on the responsible use and management of natural resources.</li> <li>Relate biodiversity to the stability and functioning of Earth systems that support human life.</li> <li>Explain how understanding the sources of environmental problems can lead to changes in human activities that reduce global impacts.</li> <li>Describe how international cooperation has reduced some large-scale environmental problems, such as acid rain and ozone depletion.</li> <li>Distinguish between environmental problems that can be mitigated through changes in human activities and those that are more difficult to reverse. (E-ESS3-4)</li> <li>Recognize that long-term sustainability involves balancing human needs with the limits of Earth’s systems. (E-ESS3-4)</li> </ul> |

## ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

| Sixth and Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify how local needs, values, and priorities shape which technologies are used in different places.</li> <li>Classify examples of technology use by regional factors such as climate, available natural resources, and economic conditions.</li> <li>Explain how a new scientific finding can shift which technology is adopted or adapted in a community.</li> <li>Analyze how a limitation (cost, materials, environmental risk) changes whether a technology is practical in the stated context.</li> </ul> <p>Seventh Grade</p> <ul style="list-style-type: none"> <li>Describe how technology use and limits on use vary with community priorities and regional conditions over time.</li> <li>Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment.</li> <li>Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out.</li> <li>Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.</li> </ul> | <ul style="list-style-type: none"> <li>Identify a societal need met by a large-scale Earth/space technology. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-4, E-ESS3-7)</li> <li>Classify options by costs, benefits, risks, and trade-offs, noting how these differ by region (climate, resources, economics). (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-4, E-ESS3-7)</li> <li>Describe how engineers refine technological systems by applying scientific research (climate data, geoscience models, natural-resource analyses) to increase benefits while reducing risks to people and ecosystems. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-4, E-ESS3-7)</li> <li>Evaluate which option is most feasible or sustainable for the stated region. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-4, E-ESS3-7)</li> <li>Identify one regional factor that most shifts the preferred choice and explain why. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-4, E-ESS3-7)</li> </ul> |

***E-ESS3-4. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.***

**ESS3.C: Human Impacts on Earth Systems**

| <b>Seventh Grade</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <ul style="list-style-type: none"><li>• Describe how human activities have altered Earth’s biosphere, including damage to or loss of natural habitats.</li><li>• Relate increases in human population and resource consumption to changes in biodiversity, ecosystem stability, and resource availability.</li><li>• Explain how human-caused environmental changes can have different effects, both negative and positive, on different living things.</li><li>• Describe examples of how monitoring environmental conditions and changing how resources are used can help reduce human impacts.</li></ul> | <ul style="list-style-type: none"><li>• Describe how advances in science and technology can reduce pollution, waste, and ecosystem degradation.</li><li>• Relate scientific understanding and technological development to changes in how humans interact with Earth systems over time.</li><li>• Distinguish between environmental problems that can be mitigated through changes in human activities and those that are more difficult to reverse. (E-ESS3-3)</li><li>• Recognize that long-term sustainability involves balancing human needs with the limits of Earth’s systems. (E-ESS3-3)</li></ul> |

## ETS1.B: Developing Possible Solutions

| Fourth and Fifth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>Fourth Grade</p> <ul style="list-style-type: none"> <li>• Create a design solution that can be used to gather evidence about how well it solves a problem.</li> <li>• Describe the likely conditions the device or solution must handle.</li> <li>• Compare proposed solutions by matching each to the stated criteria and constraints and noting any mismatch.</li> <li>• Explain which feature(s) most support better performance using grade-appropriate evidence.</li> <li>• Evaluate and communicate one targeted change that would strengthen the next version while staying within the same limits.</li> </ul> <p>Fifth Grade</p> <ul style="list-style-type: none"> <li>• Create a design solution that can be used to gather evidence about how well it solves a problem.</li> <li>• Identify important information about the stated problem from provided sources and name the specific need the solution must address.</li> <li>• Compare proposed solutions by aligning how each meets the given criteria and constraints across the likely conditions.</li> <li>• Identify which criterion (effectiveness, cost, environmental impact) should take top priority for this problem and explain why.</li> <li>• Explain which solution best fits the high-priority criteria qualitatively.</li> <li>• Evaluate one targeted revision that would improve performance in more than one condition while staying within the constraints.</li> </ul> | <ul style="list-style-type: none"> <li>• Identify criteria and constraints for the Earth-resource or impact-mitigation task, including cost, safety, reliability, aesthetics, social/cultural/environmental effects, and environmental stability. (E-ESS3-2)</li> <li>• Compare options by classifying benefits and limitations across economic, environmental, and social dimensions for the same need. (E-ESS3-2)</li> <li>• Explain which criterion should lead the decision and name one key trade-off that is accepted. (E-ESS3-2)</li> <li>• Evaluate a solution refinement that reduces an unintended impact. (E-ESS3-2)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>• Recognize when combining specific features from two options improves fit to the local context and name the pair to keep. (E-ESS3-2)</li> </ul> |

## ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

| Sixth and Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify how local needs, values, and priorities shape which technologies are used in different places.</li> <li>Classify examples of technology use by regional factors such as climate, available natural resources, and economic conditions.</li> <li>Explain how a new scientific finding can shift which technology is adopted or adapted in a community.</li> <li>Analyze how a limitation (cost, materials, environmental risk) changes whether a technology is practical in the stated context.</li> </ul> <p>Seventh Grade</p> <ul style="list-style-type: none"> <li>Describe how technology use and limits on use vary with community priorities and regional conditions over time.</li> <li>Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment.</li> <li>Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out.</li> <li>Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.</li> </ul> | <ul style="list-style-type: none"> <li>Identify a societal need met by a large-scale Earth/space technology. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-7)</li> <li>Classify options by costs, benefits, risks, and trade-offs, noting how these differ by region (climate, resources, economics). (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-7)</li> <li>Describe how engineers refine technological systems by applying scientific research (climate data, geoscience models, natural-resource analyses) to increase benefits while reducing risks to people and ecosystems. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-7)</li> <li>Evaluate which option is most feasible or sustainable for the stated region. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-7)</li> <li>Identify one regional factor that most shifts the preferred choice and explain why. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-7)</li> </ul> |

***E-ESS3-5. Analyze data and the results from global climate models to make an evidence-based forecast of the current rate of regional or global climate change and associated future impacts to Earth’s systems.***

**ESS3.D: Global Climate Change**

| <b>Seventh Grade</b>                                                                                                                                                                                                                                                                                                                                                                                         | <b>Earth and Space Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <ul style="list-style-type: none"> <li>Analyze Earth’s average surface temperature over the past century.</li> <li>Explain how interactions among human activities, the atmosphere, and other Earth systems contribute to observed changes in climate patterns.</li> <li>Use evidence to clarify the factors that have caused recent global temperature increases (for example greenhouse gases).</li> </ul> | <ul style="list-style-type: none"> <li>Distinguish between global climate trends and regional variability in climate impacts. (E-ESS3-6, E-ESS3-7)</li> <li>Use evidence from observations, historical records, and climate models to describe recent changes in global and regional climate. (E-ESS3-6, E-ESS3-7)</li> <li>Describe how recent changes in climate patterns differ in rate and magnitude from many past changes in climate patterns. (E-ESS3-6, E-ESS3-7)</li> <li>Describe how the impacts of long-term changes in climate patterns vary by region, ecosystem, and human population. (E-ESS3-6, E-ESS3-7)</li> </ul> |

***E-ESS3-6. Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity.***

ESS2.D: Weather and Climate

| Sixth Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <ul style="list-style-type: none"> <li>• Relate Earth’s axial tilt to seasonal patterns of uneven heating and cooling across the planet.</li> <li>• Explain how unequal heating of Earth contributes to large-scale patterns of weather and climate.</li> <li>• Relate interactions among sunlight, the ocean, the atmosphere, ice, landforms, and living things to differences in weather and climate patterns.</li> <li>• Describe how latitude, altitude, and geographic features influence atmospheric and oceanic circulation patterns.</li> <li>• Relate the ocean’s ability to absorb, store, and redistribute energy to its influence on regional weather and climate.</li> </ul> | <ul style="list-style-type: none"> <li>• Relate long-term atmospheric patterns to biological processes and factors that influence the composition of the atmosphere. (E-ESS2-4, E-ESS2-6, E-ESS2-7)</li> <li>• Explain how increases in greenhouse gases affect Earth’s capacity to retain energy. (E-ESS2-4, E-ESS2-6, E-ESS2-7)</li> <li>• Describe how future climate patterns are influenced by interactions among Earth systems as well as human activities. (E-ESS2-4, E-ESS2-6, E-ESS2-7)</li> <li>• Identify that feedbacks among Earth systems can amplify or reduce changes in climate patterns. (E-ESS2-4, E-ESS2-6, E-ESS2-7)</li> </ul> <p><i>Extended Knowledge</i></p> <ul style="list-style-type: none"> <li>• Recognize that long-term changes in climate patterns occur over a wide range of timescales, from years to millions of years. (E-ESS2-4, E-ESS2-6, E-ESS2-7)</li> </ul> |

## ESS3.D: Global Climate Change

| Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <ul style="list-style-type: none"> <li>Analyze Earth’s average surface temperature over the past century.</li> <li>Explain how interactions among human activities, the atmosphere, and other Earth systems contribute to observed changes in climate patterns.</li> <li>Use evidence to clarify the factors that have caused recent global temperature increases (for example greenhouse gases).</li> </ul> | <ul style="list-style-type: none"> <li>Distinguish between global climate trends and regional variability in climate impacts. (E-ESS3-5, E-ESS3-7)</li> <li>Explain global changes in climate patterns as a complex Earth-system phenomenon that occurs over long periods of Earth’s history. (E-ESS3-7)</li> <li>Relate interactions among the hydrosphere, atmosphere, geosphere, and biosphere to patterns and changes in Earth’s climate system. (E-ESS3-7)</li> <li>Use evidence from observations, historical records, and climate models to describe recent changes in global and regional climate. (E-ESS3-5, E-ESS3-7)</li> <li>Describe how recent changes in climate patterns differ in rate and magnitude from many past changes in climate patterns. (E-ESS3-5, E-ESS3-7)</li> <li>Describe how the impacts of long-term changes in climate patterns vary by region, ecosystem, and human population. (E-ESS3-5, E-ESS3-7)</li> </ul> |

## ESS3.B: Natural Hazards

| Sixth Grade                                                                                                                                                                                                                                                                                                                     | Earth and Space Science                                                                                                                                                           |
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| <ul style="list-style-type: none"> <li>Use patterns from the history of natural hazards in a region to explain where similar events are more likely to occur in the future.</li> <li>Explain how Earth processes associated with natural hazards influence the locations and likelihoods of different hazard events.</li> </ul> | <ul style="list-style-type: none"> <li>Describe how some human activities can increase the frequency or intensity of certain natural hazards, thereby increasing risk.</li> </ul> |

***E-ESS3-7. Create an argument, based on evidence that describes how changes in climate on Earth have affected human activity.***

ESS3.D: Global Climate Change

| Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <ul style="list-style-type: none"> <li>Analyze Earth’s average surface temperature over the past century.</li> <li>Explain how interactions among human activities, the atmosphere, and other Earth systems contribute to observed changes in climate patterns.</li> <li>Use evidence to clarify the factors that have caused recent global temperature increases (for example greenhouse gases).</li> <li>Recognize that scientific understanding of long-term changes in climate patterns is based on multiple sources of evidence and continues to develop over time.</li> </ul> | <ul style="list-style-type: none"> <li>Distinguish between global climate trends and regional variability in climate impacts. (E-ESS3-5, E-ESS3-6)</li> <li>Explain global changes in climate patterns as a complex Earth-system phenomenon that occurs over long periods of Earth’s history. (E-ESS3-6)</li> <li>Relate interactions among the hydrosphere, atmosphere, geosphere, and biosphere to patterns and changes in Earth’s climate system. (E-ESS3-6)</li> <li>Describe how recent changes in climate patterns differ in rate and magnitude from many past changes in climate patterns. (E-ESS3-5, E-ESS3-6)</li> <li>Describe how the impacts of long-term changes in climate patterns vary by region, ecosystem, and human population. (E-ESS3-5, E-ESS3-6)</li> <li>Explain how science-based climate models help anticipate possible future climate patterns and impacts.</li> </ul> |

## ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

| Sixth and Seventh Grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Earth and Space Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>Sixth Grade</p> <ul style="list-style-type: none"> <li>Identify how local needs, values, and priorities shape which technologies are used in different places.</li> <li>Classify examples of technology use by regional factors such as climate, available natural resources, and economic conditions.</li> <li>Explain how a new scientific finding can shift which technology is adopted or adapted in a community.</li> <li>Analyze how a limitation (cost, materials, environmental risk) changes whether a technology is practical in the stated context.</li> </ul> <p>Seventh Grade</p> <ul style="list-style-type: none"> <li>Describe how technology use and limits on use vary with community priorities and regional conditions over time.</li> <li>Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment.</li> <li>Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out.</li> <li>Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.</li> </ul> | <ul style="list-style-type: none"> <li>Identify a societal need met by a large-scale Earth/space technology. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4)</li> <li>Classify options by costs, benefits, risks, and trade-offs, noting how these differ by region (climate, resources, economics). (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4)</li> <li>Describe how engineers refine technological systems by applying scientific research (climate data, geoscience models, natural-resource analyses) to increase benefits while reducing risks to people and ecosystems. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4)</li> <li>Evaluate which option is most feasible or sustainable for the stated region. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4)</li> <li>Identify one regional factor that most shifts the preferred choice and explain why. (E-ESS2-2 E-ESS3-1, E-ESS3-2, E-ESS3-3, E-ESS3-4)</li> </ul> |